

Machine Learning

```
# Load the dataset
df <- read_csv("train.csv")

## Rows: 26570 Columns: 26
## — Column specification


---


## Delimiter: ","
## chr (4): product_code, attribute_0, attribute_1, failure
## dbl (22): id, loading, attribute_2, attribute_3, measurement_0,
measurement_...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.

# Basic structure and summary of the dataset
str(df)

## spc_tbl_ [26,570 × 26] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ id : num [1:26570] 0 1 2 3 4 5 6 7 8 9 ...
## $ product_code : chr [1:26570] "A" "A" "A" "A" ...
## $ loading : num [1:26570] 80.1 84.9 82.4 101.1 188.1 ...
## $ attribute_0 : chr [1:26570] "material_7" "material_7" "material_7"
"material_7" ...
## $ attribute_1 : chr [1:26570] "material_8" "material_8" "material_8"
"material_8" ...
## $ attribute_2 : num [1:26570] 9 9 9 9 9 9 9 9 9 9 ...
## $ attribute_3 : num [1:26570] 5 5 5 5 5 5 5 5 5 5 ...
## $ measurement_0 : num [1:26570] 7 14 12 13 9 11 12 4 9 10 ...
## $ measurement_1 : num [1:26570] 8 3 1 2 2 4 2 8 6 4 ...
## $ measurement_2 : num [1:26570] 4 3 5 6 8 0 4 8 5 7 ...
## $ measurement_3 : num [1:26570] 18 18.2 18.1 17.3 19.3 ...
## $ measurement_4 : num [1:26570] 12.5 11.5 11.7 11.2 12.9 ...
## $ measurement_5 : num [1:26570] 15.7 17.7 16.7 18.6 17 ...
## $ measurement_6 : num [1:26570] 19.3 17.9 18.2 18.3 15.7 ...
## $ measurement_7 : num [1:26570] 11.7 12.7 12.7 12.6 11.3 ...
## $ measurement_8 : num [1:26570] 20.2 17.9 18.3 19.1 18.1 ...
## $ measurement_9 : num [1:26570] 10.7 12.4 12.7 12.5 10.3 ...
## $ measurement_10: num [1:26570] 15.9 17.9 15.6 16.3 17.1 ...
## $ measurement_11: num [1:26570] 17.6 17.9 NA 18.4 19.9 ...
## $ measurement_12: num [1:26570] 15.2 11.8 13.8 10 12.4 ...
## $ measurement_13: num [1:26570] 15 14.7 16.7 15.2 16.2 ...
## $ measurement_14: num [1:26570] NA 15.4 18.6 15.6 12.8 ...
## $ measurement_15: num [1:26570] 13 14.4 14.1 16.2 13.2 ...
## $ measurement_16: num [1:26570] 14.7 15.6 17.9 17.2 16.4 ...
## $ measurement_17: num [1:26570] 764 682 663 826 580 ...
```

```
## $ failure      : chr [1:26570] "No" "No" "No" "No" ...
## - attr(*, "spec")=
## .. cols(
## ..   id = col_double(),
## ..   product_code = col_character(),
## ..   loading = col_double(),
## ..   attribute_0 = col_character(),
## ..   attribute_1 = col_character(),
## ..   attribute_2 = col_double(),
## ..   attribute_3 = col_double(),
## ..   measurement_0 = col_double(),
## ..   measurement_1 = col_double(),
## ..   measurement_2 = col_double(),
## ..   measurement_3 = col_double(),
## ..   measurement_4 = col_double(),
## ..   measurement_5 = col_double(),
## ..   measurement_6 = col_double(),
## ..   measurement_7 = col_double(),
## ..   measurement_8 = col_double(),
## ..   measurement_9 = col_double(),
## ..   measurement_10 = col_double(),
## ..   measurement_11 = col_double(),
## ..   measurement_12 = col_double(),
## ..   measurement_13 = col_double(),
## ..   measurement_14 = col_double(),
## ..   measurement_15 = col_double(),
## ..   measurement_16 = col_double(),
## ..   measurement_17 = col_double(),
## ..   failure = col_character()
## .. )
## - attr(*, "problems")=<externalptr>
```

summary(df)

```
##      id      product_code      loading      attribute_0
## Min.   :    0  Length:26570  Min.   : 33.16  Length:26570
## 1st Qu.: 6642  Class :character 1st Qu.: 99.99  Class :character
## Median :13284  Mode  :character Median :122.39  Mode  :character
## Mean   :13284  Mean   :127.83
## 3rd Qu.:19927  3rd Qu.:149.15
## Max.   :26569  Max.   :385.86
##              NA's   :250
## attribute_1      attribute_2      attribute_3      measurement_0
## Length:26570      Min.   :5.000  Min.   :5.00  Min.   : 0.000
## Class :character  1st Qu.:6.000  1st Qu.:6.00  1st Qu.: 4.000
## Mode  :character  Median :6.000  Median :8.00  Median : 7.000
##              Mean   :6.754  Mean   :7.24  Mean   : 7.416
##              3rd Qu.:8.000  3rd Qu.:8.00  3rd Qu.:10.000
##              Max.   :9.000  Max.   :9.00  Max.   :29.000
##
```

```
## measurement_1 measurement_2 measurement_3 measurement_4
## Min. : 0.000 Min. : 0.000 Min. :13.97 Min. : 8.008
## 1st Qu.: 5.000 1st Qu.: 4.000 1st Qu.:17.12 1st Qu.:11.051
## Median : 8.000 Median : 6.000 Median :17.79 Median :11.733
## Mean : 8.233 Mean : 6.257 Mean :17.79 Mean :11.732
## 3rd Qu.:11.000 3rd Qu.: 8.000 3rd Qu.:18.47 3rd Qu.:12.410
## Max. :29.000 Max. :24.000 Max. :21.50 Max. :16.484
## NA's :381 NA's :538
## measurement_5 measurement_6 measurement_7 measurement_8
## Min. :12.07 Min. :12.71 Min. : 7.968 Min. :15.22
## 1st Qu.:16.44 1st Qu.:16.84 1st Qu.:11.045 1st Qu.:18.34
## Median :17.13 Median :17.52 Median :11.712 Median :19.02
## Mean :17.13 Mean :17.51 Mean :11.717 Mean :19.02
## 3rd Qu.:17.80 3rd Qu.:18.18 3rd Qu.:12.391 3rd Qu.:19.71
## Max. :21.43 Max. :21.54 Max. :15.419 Max. :23.81
## NA's :676 NA's :796 NA's :937 NA's :1048
## measurement_9 measurement_10 measurement_11 measurement_12
## Min. : 7.537 Min. : 9.323 Min. :12.46 Min. : 5.167
## 1st Qu.:10.757 1st Qu.:15.209 1st Qu.:18.17 1st Qu.:10.703
## Median :11.430 Median :16.127 Median :19.21 Median :11.717
## Mean :11.431 Mean :16.118 Mean :19.17 Mean :11.703
## 3rd Qu.:12.102 3rd Qu.:17.025 3rd Qu.:20.21 3rd Qu.:12.709
## Max. :15.412 Max. :22.479 Max. :25.64 Max. :17.663
## NA's :1227 NA's :1300 NA's :1468 NA's :1601
## measurement_13 measurement_14 measurement_15 measurement_16
## Min. :10.89 Min. : 9.14 Min. : 9.104 Min. : 9.701
## 1st Qu.:14.89 1st Qu.:15.06 1st Qu.:13.957 1st Qu.:15.268
## Median :15.63 Median :16.04 Median :14.969 Median :16.436
## Mean :15.65 Mean :16.05 Mean :14.996 Mean :16.461
## 3rd Qu.:16.37 3rd Qu.:17.08 3rd Qu.:16.018 3rd Qu.:17.628
## Max. :22.71 Max. :22.30 Max. :21.626 Max. :24.094
## NA's :1774 NA's :1874 NA's :2009 NA's :2110
## measurement_17 failure
## Min. : 196.8 Length:26570
## 1st Qu.: 619.0 Class :character
## Median : 701.0 Mode :character
## Mean : 701.3
## 3rd Qu.: 784.1
## Max. :1312.8
## NA's :2284
```

Check for missing values

```
missing_values <- colSums(is.na(df))
```

```
print(missing_values)
```

```
## id product_code loading attribute_0 attribute_1
## 0 0 250 0 0
## attribute_2 attribute_3 measurement_0 measurement_1 measurement_2
## 0 0 0 0 0
```

```
## measurement_3 measurement_4 measurement_5 measurement_6 measurement_7
##           381           538           676           796           937
## measurement_8 measurement_9 measurement_10 measurement_11 measurement_12
##          1048          1227          1300          1468          1601
## measurement_13 measurement_14 measurement_15 measurement_16 measurement_17
##          1774          1874          2009          2110          2284
##           failure
##              0
```

Check for missing values

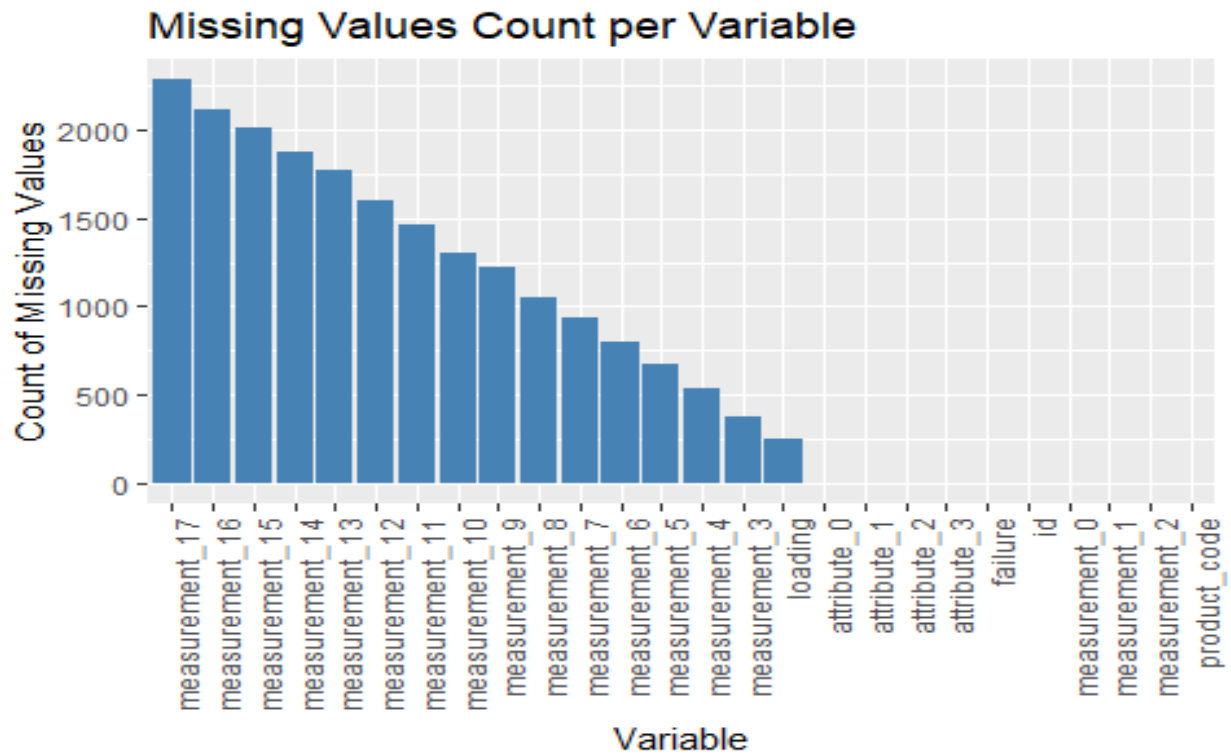
```
missing_values <- colSums(is.na(df))
print(missing_values)
```

```
##           id product_code loading attribute_0 attribute_1
##           0           0      250           0           0
## attribute_2 attribute_3 measurement_0 measurement_1 measurement_2
##           0           0           0           0           0
## measurement_3 measurement_4 measurement_5 measurement_6 measurement_7
##           381           538           676           796           937
## measurement_8 measurement_9 measurement_10 measurement_11 measurement_12
##          1048          1227          1300          1468          1601
## measurement_13 measurement_14 measurement_15 measurement_16 measurement_17
##          1774          1874          2009          2110          2284
##           failure
##              0
```

Visualize missing values

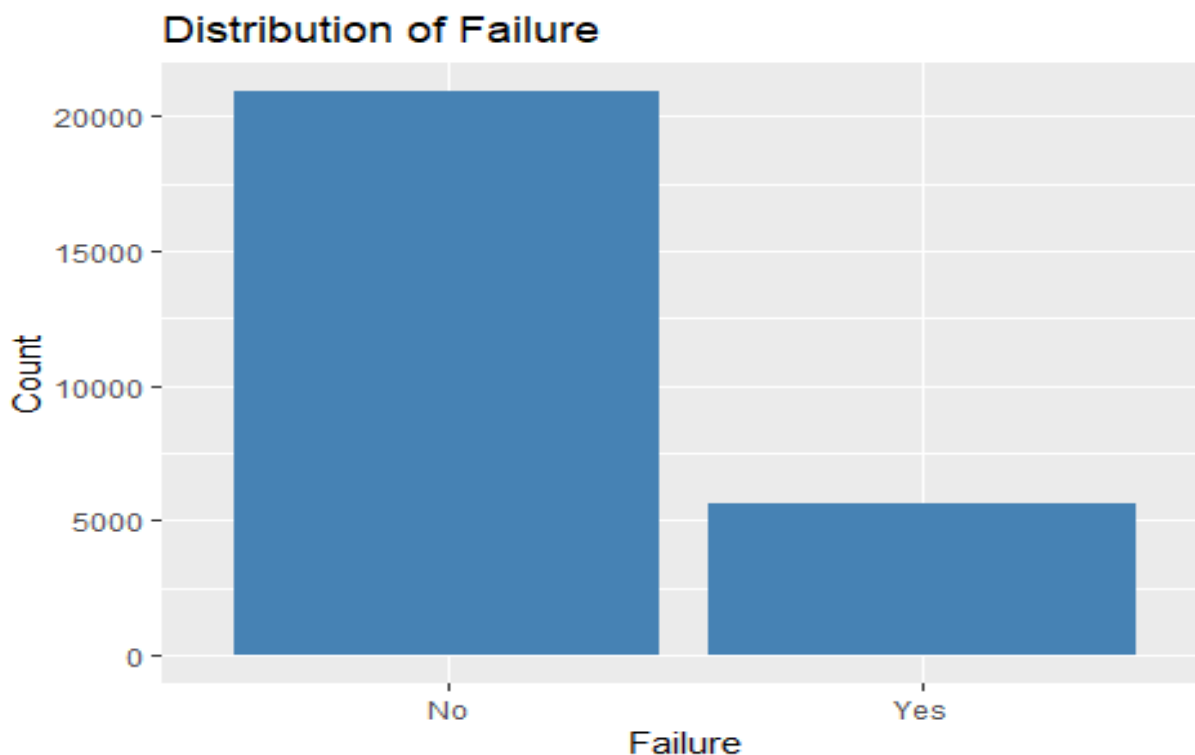
```
missing_values_df <- data.frame(
  variable = names(missing_values),
  missing_count = missing_values
)
```

```
ggplot(missing_values_df, aes(x = reorder(variable, -missing_count), y =
missing_count)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(title = "Missing Values Count per Variable", x = "Variable", y =
"Count of Missing Values")
```



Distribution of the target variable

```
ggplot(df, aes(x = failure)) +  
  geom_bar(fill = "steelblue") +  
  labs(title = "Distribution of Failure", x = "Failure", y = "Count")
```



```
# Summary statistics of numerical features
num_features <- df %>% select_if(is.numeric)
```

```
summary(num_features)
```

```
##      id      loading      attribute_2      attribute_3
## Min.   :    0   Min.   : 33.16   Min.   :5.000   Min.   :5.00
## 1st Qu.: 6642   1st Qu.: 99.99   1st Qu.:6.000   1st Qu.:6.00
## Median :13284   Median :122.39   Median :6.000   Median :8.00
## Mean   :13284   Mean   :127.83   Mean   :6.754   Mean   :7.24
## 3rd Qu.:19927   3rd Qu.:149.15   3rd Qu.:8.000   3rd Qu.:8.00
## Max.   :26569   Max.   :385.86   Max.   :9.000   Max.   :9.00
##
##      NA's      :250
## measurement_0 measurement_1 measurement_2 measurement_3
## Min.   : 0.000   Min.   : 0.000   Min.   : 0.000   Min.   :13.97
## 1st Qu.: 4.000   1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:17.12
## Median : 7.000   Median : 8.000   Median : 6.000   Median :17.79
## Mean   : 7.416   Mean   : 8.233   Mean   : 6.257   Mean   :17.79
## 3rd Qu.:10.000   3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:18.47
## Max.   :29.000   Max.   :29.000   Max.   :24.000   Max.   :21.50
##
##      NA's      :381
## measurement_4 measurement_5 measurement_6 measurement_7
## Min.   : 8.008   Min.   :12.07   Min.   :12.71   Min.   : 7.968
## 1st Qu.:11.051   1st Qu.:16.44   1st Qu.:16.84   1st Qu.:11.045
## Median :11.733   Median :17.13   Median :17.52   Median :11.712
## Mean   :11.732   Mean   :17.13   Mean   :17.51   Mean   :11.717
```

```
## 3rd Qu.:12.410 3rd Qu.:17.80 3rd Qu.:18.18 3rd Qu.:12.391
## Max. :16.484 Max. :21.43 Max. :21.54 Max. :15.419
## NA's :538 NA's :676 NA's :796 NA's :937
## measurement_8 measurement_9 measurement_10 measurement_11
## Min. :15.22 Min. : 7.537 Min. : 9.323 Min. :12.46
## 1st Qu.:18.34 1st Qu.:10.757 1st Qu.:15.209 1st Qu.:18.17
## Median :19.02 Median :11.430 Median :16.127 Median :19.21
## Mean :19.02 Mean :11.431 Mean :16.118 Mean :19.17
## 3rd Qu.:19.71 3rd Qu.:12.102 3rd Qu.:17.025 3rd Qu.:20.21
## Max. :23.81 Max. :15.412 Max. :22.479 Max. :25.64
## NA's :1048 NA's :1227 NA's :1300 NA's :1468
## measurement_12 measurement_13 measurement_14 measurement_15
## Min. : 5.167 Min. :10.89 Min. : 9.14 Min. : 9.104
## 1st Qu.:10.703 1st Qu.:14.89 1st Qu.:15.06 1st Qu.:13.957
## Median :11.717 Median :15.63 Median :16.04 Median :14.969
## Mean :11.703 Mean :15.65 Mean :16.05 Mean :14.996
## 3rd Qu.:12.709 3rd Qu.:16.37 3rd Qu.:17.08 3rd Qu.:16.018
## Max. :17.663 Max. :22.71 Max. :22.30 Max. :21.626
## NA's :1601 NA's :1774 NA's :1874 NA's :2009
## measurement_16 measurement_17
## Min. : 9.701 Min. : 196.8
## 1st Qu.:15.268 1st Qu.: 619.0
## Median :16.436 Median : 701.0
## Mean :16.461 Mean : 701.3
## 3rd Qu.:17.628 3rd Qu.: 784.1
## Max. :24.094 Max. :1312.8
## NA's :2110 NA's :2284
```

Pairwise correlation of numerical features

```
correlation_matrix <- cor(num_features, use = "complete.obs")
```

```
print(correlation_matrix)
```

```
##          id      loading  attribute_2  attribute_3
## id      1.000000000 -0.001227611 -0.7416717204 0.5588366339
## loading -0.001227611 1.000000000 -0.0042578083 -0.0064238120
## attribute_2 -0.741671720 -0.004257808 1.0000000000 -0.5424164596
## attribute_3 0.558836634 -0.006423812 -0.5424164596 1.0000000000
## measurement_0 -0.260232516 -0.006281351 0.1845189637 -0.3712635287
## measurement_1 0.411263955 -0.004865948 -0.3896409435 0.3339290716
## measurement_2 0.002982388 -0.010419286 0.0239669548 0.0235629536
## measurement_3 -0.003570652 -0.007615697 0.0008194619 -0.0023157225
## measurement_4 0.013575970 -0.022110459 -0.0136728348 0.0110547345
## measurement_5 -0.012175785 0.015019826 0.0078962406 -0.0045442618
## measurement_6 0.009540738 0.008485838 -0.0048941678 -0.0002651857
## measurement_7 -0.005300416 -0.007798602 0.0014623612 -0.0089800385
## measurement_8 0.007368236 0.002890764 -0.0063937434 -0.0048118807
## measurement_9 -0.001304807 -0.007094734 0.0009507014 -0.0176354968
## measurement_10 -0.029781398 0.003617095 -0.0037218302 -0.0926208872
## measurement_11 0.007699258 0.003144806 0.0972826340 -0.0240318118
```

```

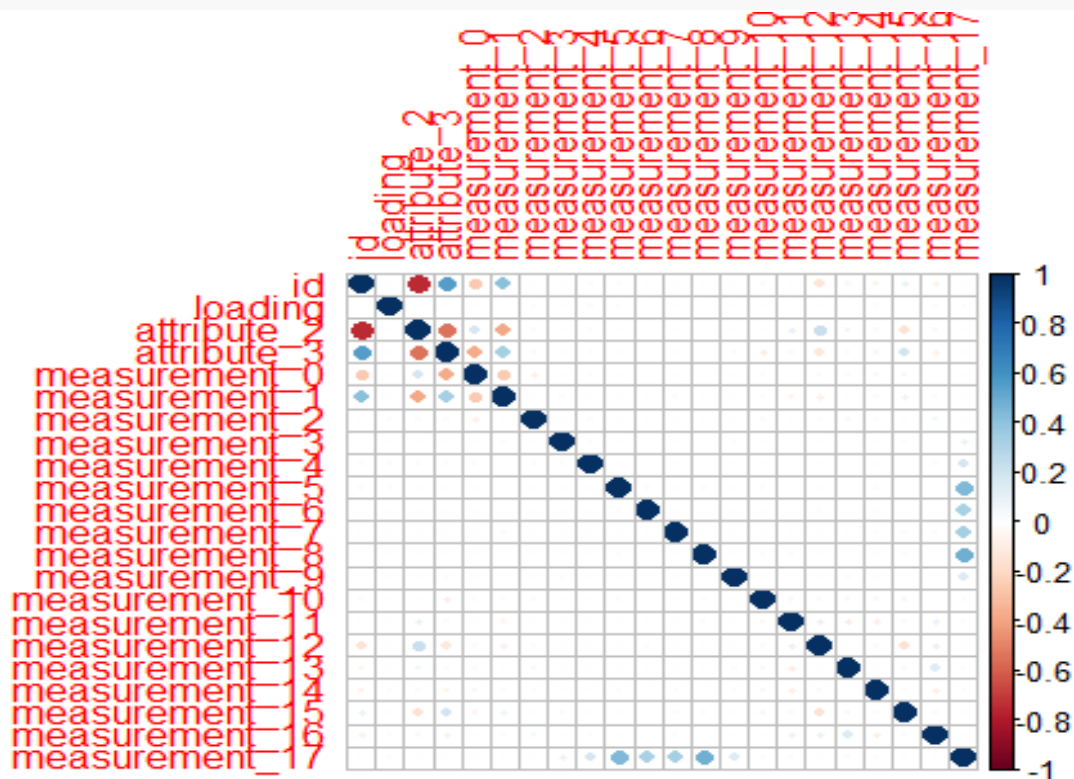
## measurement_12 -0.149744775 -0.003047974 0.2269214082 -0.1252671446
## measurement_13 -0.040824010 0.002607303 0.0559917385 0.0189812803
## measurement_14 -0.060201279 0.011127349 0.0241604359 -0.0575019063
## measurement_15 0.080293420 -0.008167758 -0.1587903172 0.1908649999
## measurement_16 -0.063092272 -0.015450909 0.0340862661 -0.0624995614
## measurement_17 0.002929289 0.014000176 -0.0032716851 -0.0012324506
## measurement_0 measurement_1 measurement_2 measurement_3
## id -0.2602325156 0.4112639548 0.0029823876 -0.0035706524
## loading -0.0062813515 -0.0048659484 -0.0104192862 -0.0076156969
## attribute_2 0.1845189637 -0.3896409435 0.0239669548 0.0008194619
## attribute_3 -0.3712635287 0.3339290716 0.0235629536 -0.0023157225
## measurement_0 1.0000000000 -0.2553509561 -0.0886333025 -0.0048561185
## measurement_1 -0.2553509561 1.0000000000 0.0067996454 -0.0101106655
## measurement_2 -0.0886333025 0.0067996454 1.0000000000 -0.0020485857
## measurement_3 -0.0048561185 -0.0101106655 -0.0020485857 1.0000000000
## measurement_4 -0.0173258115 0.0051940716 -0.0127023189 0.0169674839
## measurement_5 0.0087304531 -0.0079846014 0.0070816389 0.0032842698
## measurement_6 0.0003465680 0.0001999684 -0.0011458018 0.0097879670
## measurement_7 0.0071460294 -0.0214424855 0.0010841841 0.0064855321
## measurement_8 -0.0005092300 0.0064650840 -0.0073741202 -0.0012562489
## measurement_9 0.0096493929 0.0042041811 0.0053184770 0.0163112694
## measurement_10 0.0172206362 0.0180105803 0.0225681691 -0.0070220251
## measurement_11 -0.0003332046 -0.0443658146 0.0013856749 -0.0077962494
## measurement_12 0.0202014278 -0.0829470094 0.0140337265 -0.0063064550
## measurement_13 -0.0211899613 -0.0032950177 0.0346321064 0.0079830267
## measurement_14 0.0328154758 -0.0228651346 0.0177737110 -0.0068853238
## measurement_15 -0.0599771015 0.0721362952 -0.0022456283 -0.0166941702
## measurement_16 0.0060327190 -0.0113728473 0.0417190430 -0.0016743814
## measurement_17 0.0047791778 -0.0087702039 -0.0007490003 0.0858403967
## measurement_4 measurement_5 measurement_6 measurement_7
## id 0.0135759703 -0.012175785 0.0095407375 -0.0053004160
## loading -0.0221104594 0.015019826 0.0084858381 -0.0077986023
## attribute_2 -0.0136728348 0.007896241 -0.0048941678 0.0014623612
## attribute_3 0.0110547345 -0.004544262 -0.0002651857 -0.0089800385
## measurement_0 -0.0173258115 0.008730453 0.0003465680 0.0071460294
## measurement_1 0.0051940716 -0.007984601 0.0001999684 -0.0214424855
## measurement_2 -0.0127023189 0.007081639 -0.0011458018 0.0010841841
## measurement_3 0.0169674839 0.003284270 0.0097879670 0.0064855321
## measurement_4 1.0000000000 0.005393420 -0.0071888786 -0.0035860073
## measurement_5 0.0053934200 1.0000000000 0.0115343571 -0.0036341704
## measurement_6 -0.0071888786 0.011534357 1.0000000000 0.0052947926
## measurement_7 -0.0035860073 -0.003634170 0.0052947926 1.0000000000
## measurement_8 -0.0002745865 -0.010024045 0.0068679507 0.0064794287
## measurement_9 0.0061438552 -0.009561867 0.0057439365 0.0017614830
## measurement_10 -0.0082325553 0.014669622 -0.0055593866 -0.0060018817
## measurement_11 -0.0056903826 0.002979151 -0.0074752059 -0.0076307159
## measurement_12 0.0047419988 -0.001400851 -0.0182731841 -0.0082258956
## measurement_13 0.0075927869 -0.009360917 -0.0070430029 0.0003398175
## measurement_14 0.0056751691 0.012611799 0.0033779574 0.0139945736
## measurement_15 0.0009054383 -0.009656903 -0.0009747800 -0.0112691905

```


## measurement_16	0.0148895891	-0.009202180	-0.0059128740	-0.0111877440
## measurement_17	0.1853432174	0.447781888	0.3249755718	0.3347025263
##	measurement_8	measurement_9	measurement_10	measurement_11
## id	0.0073682364	-0.0013048067	-0.029781398	0.0076992579
## loading	0.0028907637	-0.0070947341	0.003617095	0.0031448059
## attribute_2	-0.0063937434	0.0009507014	-0.003721830	0.0972826340
## attribute_3	-0.0048118807	-0.0176354968	-0.092620887	-0.0240318118
## measurement_0	-0.0005092300	0.0096493929	0.017220636	-0.0003332046
## measurement_1	0.0064650840	0.0042041811	0.018010580	-0.0443658146
## measurement_2	-0.0073741202	0.0053184770	0.022568169	0.0013856749
## measurement_3	-0.0012562489	0.0163112694	-0.007022025	-0.0077962494
## measurement_4	-0.0002745865	0.0061438552	-0.008232555	-0.0056903826
## measurement_5	-0.0100240448	-0.0095618666	0.014669622	0.0029791509
## measurement_6	0.0068679507	0.0057439365	-0.005559387	-0.0074752059
## measurement_7	0.0064794287	0.0017614830	-0.006001882	-0.0076307159
## measurement_8	1.0000000000	-0.0030214834	0.018600300	-0.0186305571
## measurement_9	-0.0030214834	1.0000000000	-0.009626847	0.0015377070
## measurement_10	0.0186003002	-0.0096268469	1.0000000000	-0.0325754203
## measurement_11	-0.0186305571	0.0015377070	-0.032575420	1.0000000000
## measurement_12	-0.0038721067	-0.0036201626	0.024138951	0.0992313646
## measurement_13	-0.0062674566	-0.0144811372	0.033340910	-0.1008978309
## measurement_14	-0.0024671293	-0.0102993948	0.038028538	-0.0858220986
## measurement_15	0.0157028769	-0.0021289729	-0.051749081	-0.0279198437
## measurement_16	0.0073199658	-0.0089166401	-0.002174862	0.0855922497
## measurement_17	0.4885411929	0.1423083084	0.012970194	-0.0177227001
##	measurement_12	measurement_13	measurement_14	measurement_15
## id	-0.149744775	-0.0408240101	-0.060201279	0.0802934203
## loading	-0.003047974	0.0026073034	0.011127349	-0.0081677581
## attribute_2	0.226921408	0.0559917385	0.024160436	-0.1587903172
## attribute_3	-0.125267145	0.0189812803	-0.057501906	0.1908649999
## measurement_0	0.020201428	-0.0211899613	0.032815476	-0.0599771015
## measurement_1	-0.082947009	-0.0032950177	-0.022865135	0.0721362952
## measurement_2	0.014033726	0.0346321064	0.017773711	-0.0022456283
## measurement_3	-0.006306455	0.0079830267	-0.006885324	-0.0166941702
## measurement_4	0.004741999	0.0075927869	0.005675169	0.0009054383
## measurement_5	-0.001400851	-0.0093609171	0.012611799	-0.0096569030
## measurement_6	-0.018273184	-0.0070430029	0.003377957	-0.0009747800
## measurement_7	-0.008225896	0.0003398175	0.013994574	-0.0112691905
## measurement_8	-0.003872107	-0.0062674566	-0.002467129	0.0157028769
## measurement_9	-0.003620163	-0.0144811372	-0.010299395	-0.0021289729
## measurement_10	0.024138951	0.0333409104	0.038028538	-0.0517490807
## measurement_11	0.099231365	-0.1008978309	-0.085822099	-0.0279198437
## measurement_12	1.0000000000	0.0085409787	-0.025417213	-0.1505869743
## measurement_13	0.008540979	1.0000000000	-0.003362890	0.0302727879
## measurement_14	-0.025417213	-0.0033628901	1.0000000000	0.0125580346
## measurement_15	-0.150586974	0.0302727879	0.012558035	1.0000000000
## measurement_16	0.099271988	0.1405199528	-0.086709367	0.0238252513
## measurement_17	-0.009104654	-0.0106383291	0.011817030	0.0038442852
##	measurement_16	measurement_17		
## id	-0.063092272	0.0029292895		

```
## loading          -0.015450909    0.0140001757
## attribute_2      0.034086266   -0.0032716851
## attribute_3     -0.062499561   -0.0012324506
## measurement_0    0.006032719    0.0047791778
## measurement_1   -0.011372847   -0.0087702039
## measurement_2    0.041719043   -0.0007490003
## measurement_3   -0.001674381    0.0858403967
## measurement_4    0.014889589    0.1853432174
## measurement_5   -0.009202180    0.4477818878
## measurement_6   -0.005912874    0.3249755718
## measurement_7   -0.011187744    0.3347025263
## measurement_8    0.007319966    0.4885411929
## measurement_9   -0.008916640    0.1423083084
## measurement_10  -0.002174862    0.0129701942
## measurement_11  0.085592250   -0.0177227001
## measurement_12  0.099271988   -0.0091046535
## measurement_13  0.140519953   -0.0106383291
## measurement_14  -0.086709367    0.0118170302
## measurement_15  0.023825251    0.0038442852
## measurement_16  1.000000000    0.0001940920
## measurement_17  0.000194092    1.0000000000
```

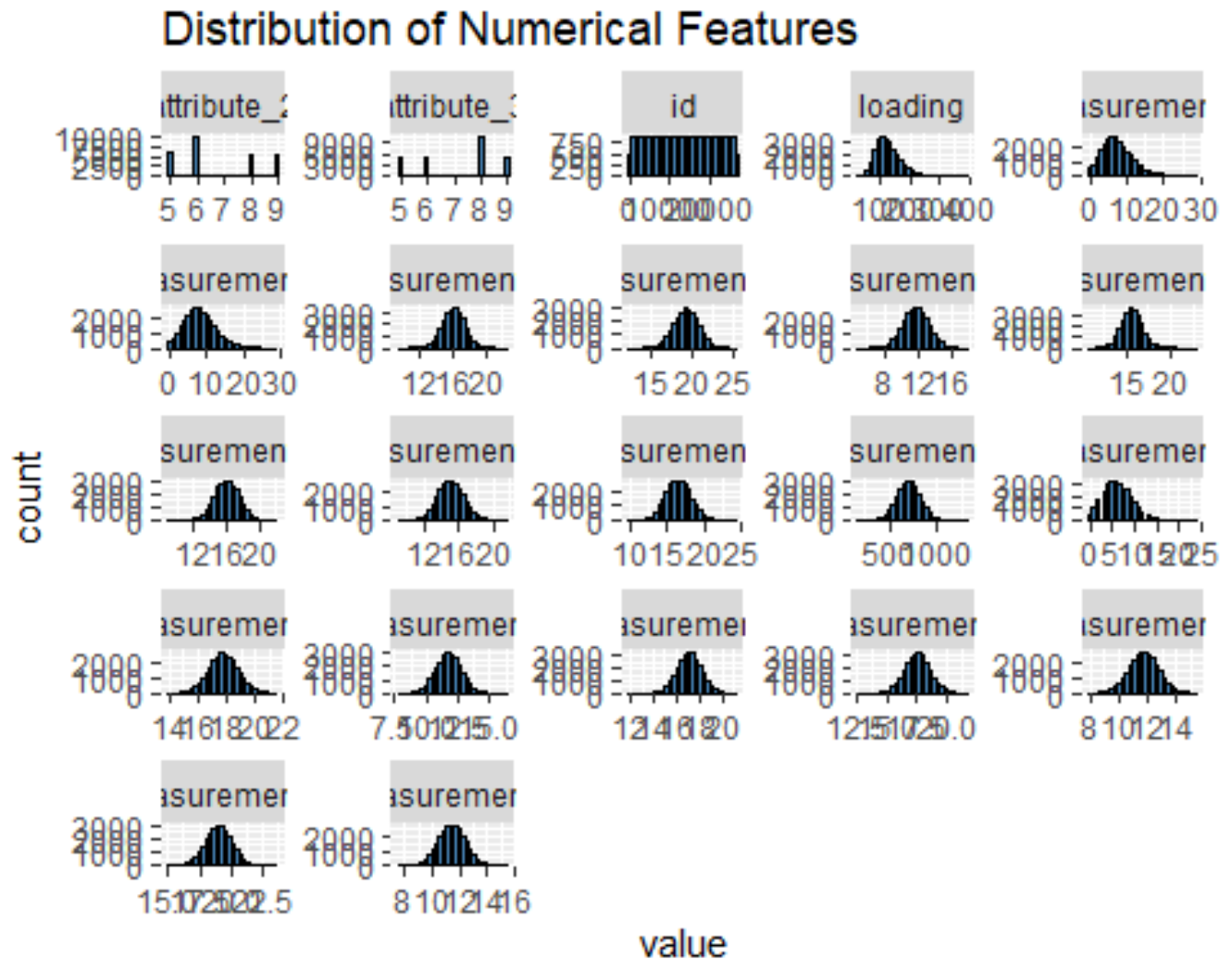
```
# Visualize the correlation matrix
corrplot(correlation_matrix, method = "circle")
```



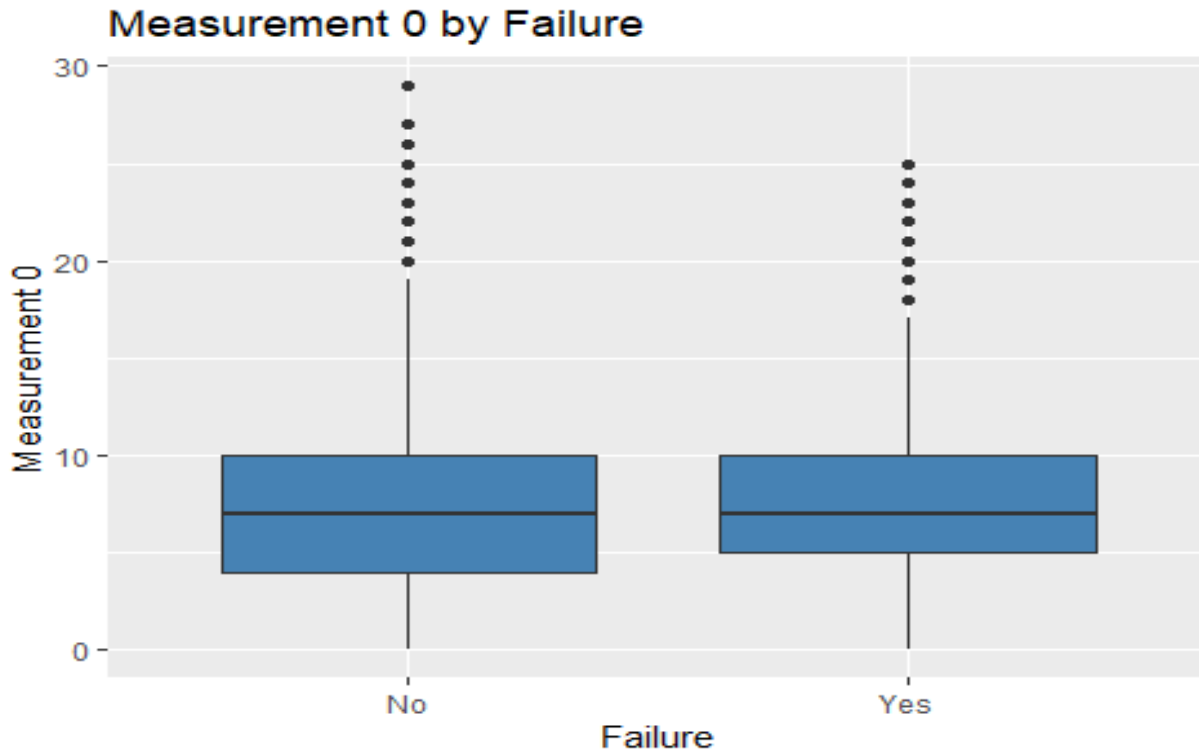
```
# Distribution of numerical features
num_features_long <- gather(num_features)
```

```
ggplot(num_features_long, aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black") +
  facet_wrap(~ key, scales = "free") +
  labs(title = "Distribution of Numerical Features")
```

```
## Warning: Removed 20273 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

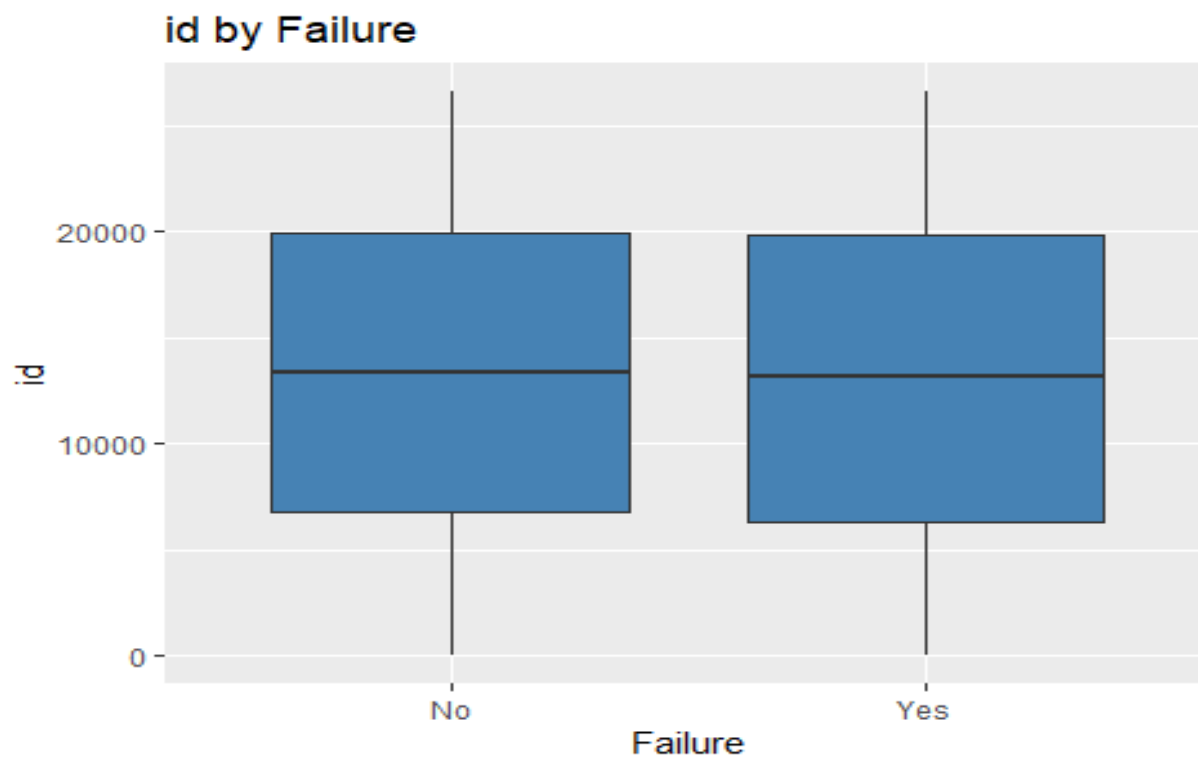


```
# Boxplots of numerical features by failure
ggplot(df, aes(x = failure, y = measurement_0)) +
  geom_boxplot(fill = "steelblue") +
  labs(title = "Measurement 0 by Failure", x = "Failure", y = "Measurement
0")
```

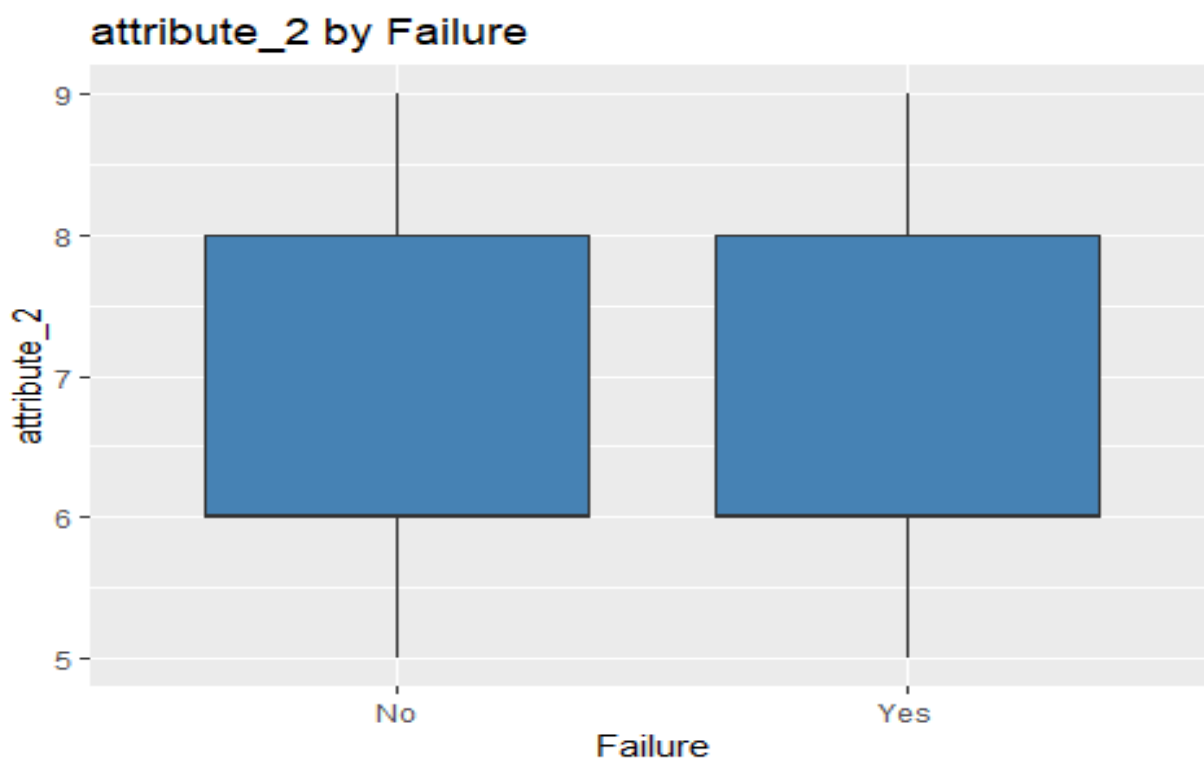
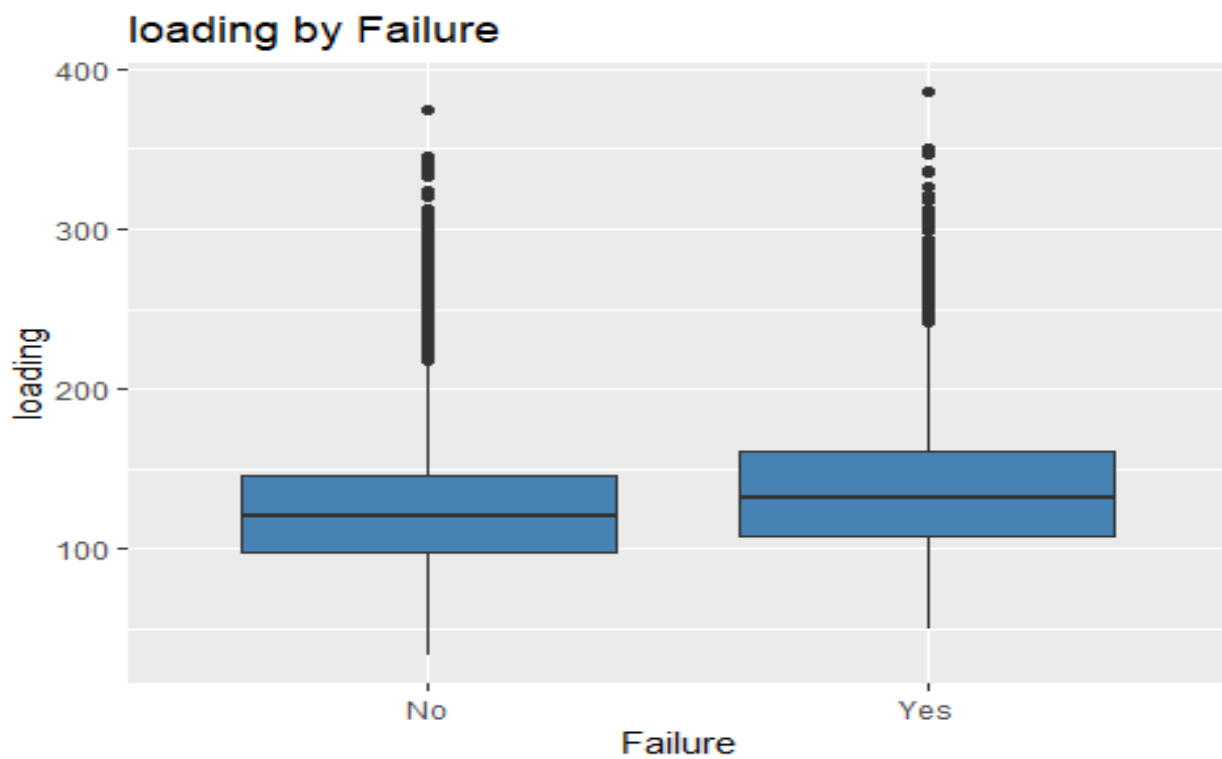


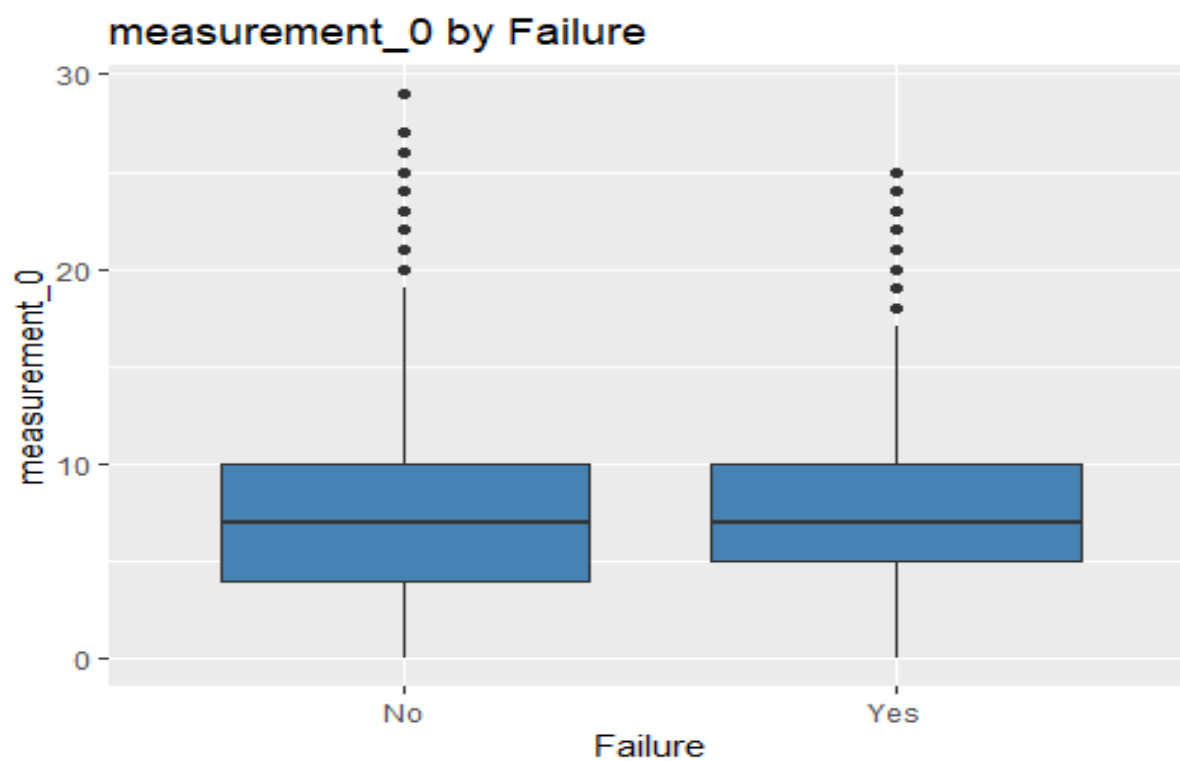
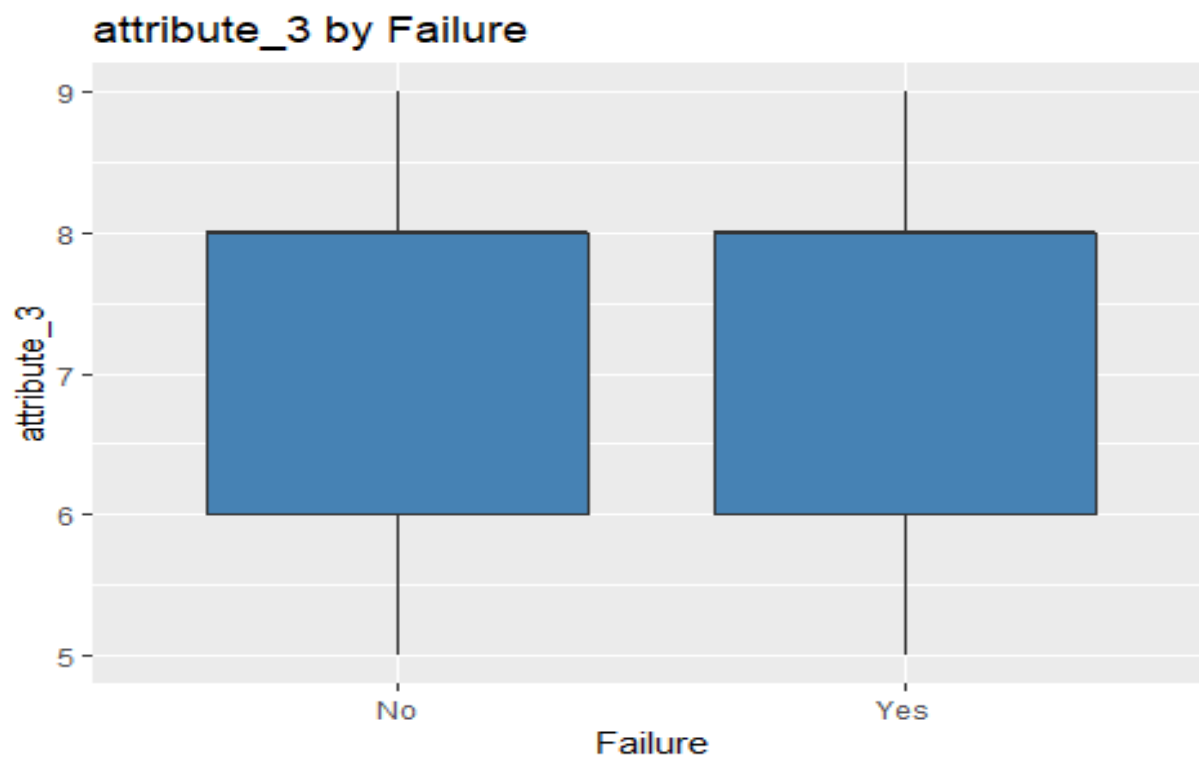
```
# Repeat for other numerical features
for (feature in names(num_features)) {
  p <- ggplot(df, aes_string(x = "failure", y = feature)) +
    geom_boxplot(fill = "steelblue") +
    labs(title = paste(feature, "by Failure"), x = "Failure", y = feature)
  print(p)
}

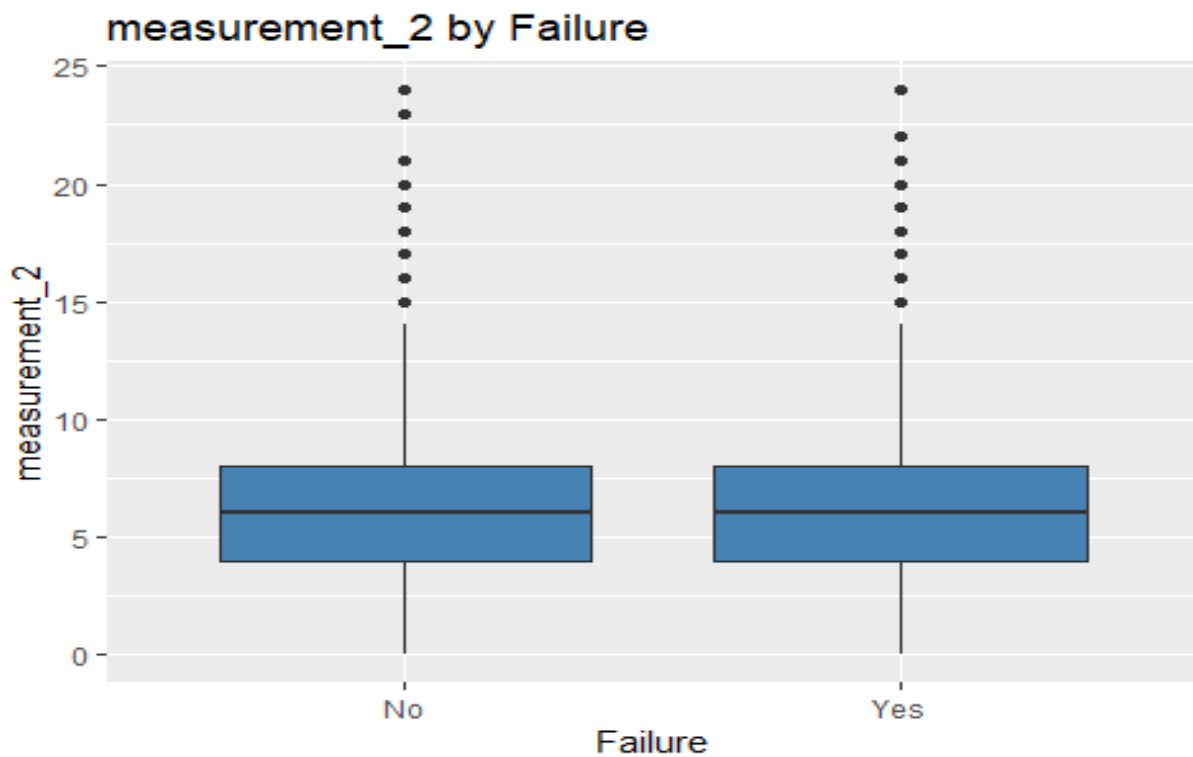
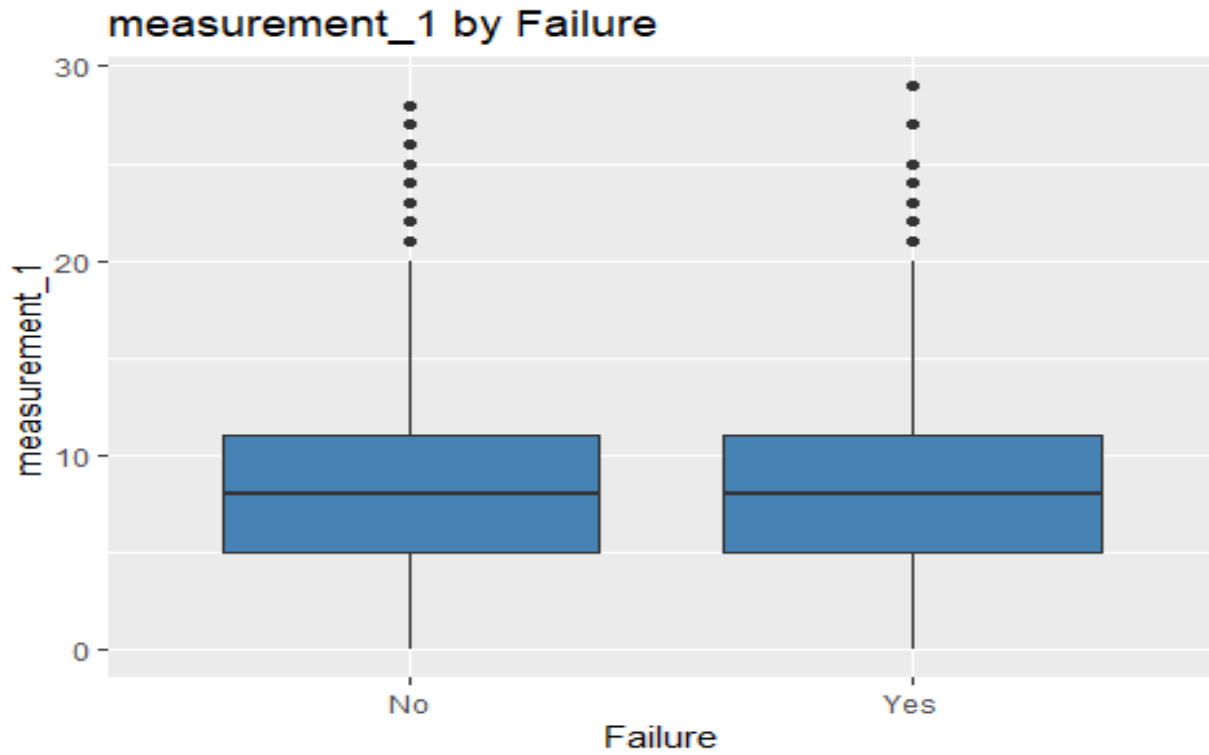
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



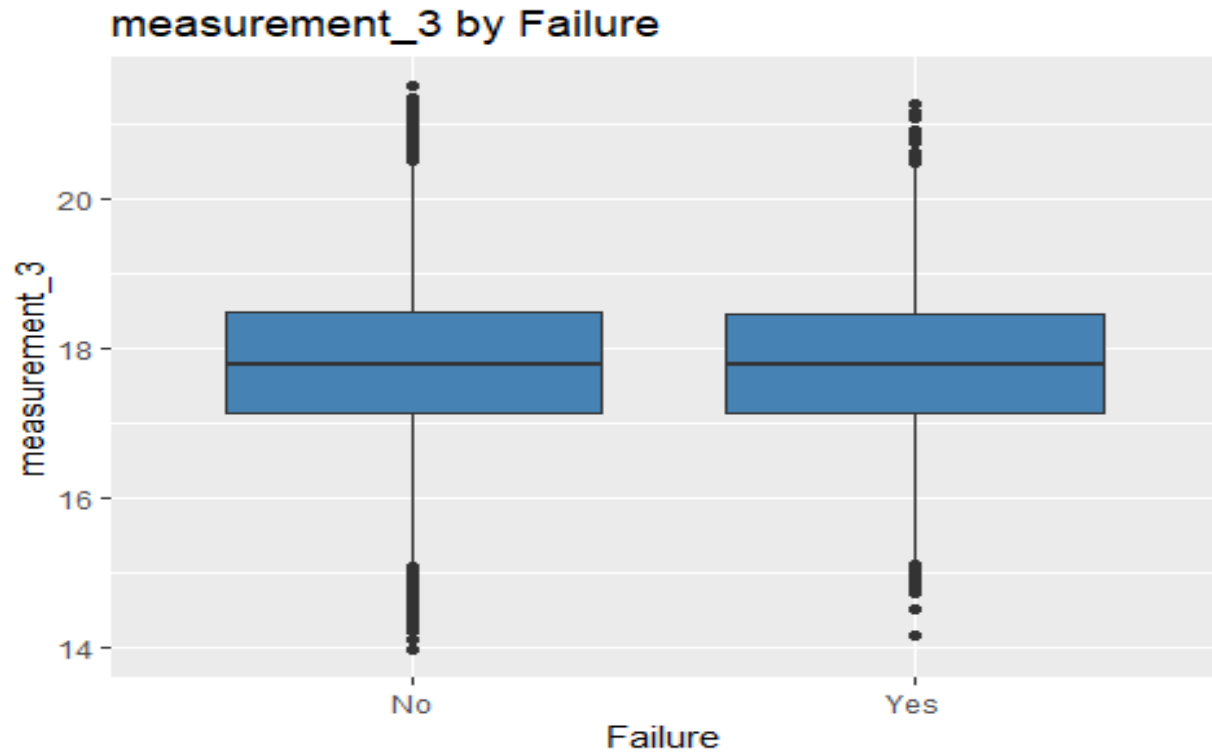
```
## Warning: Removed 250 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



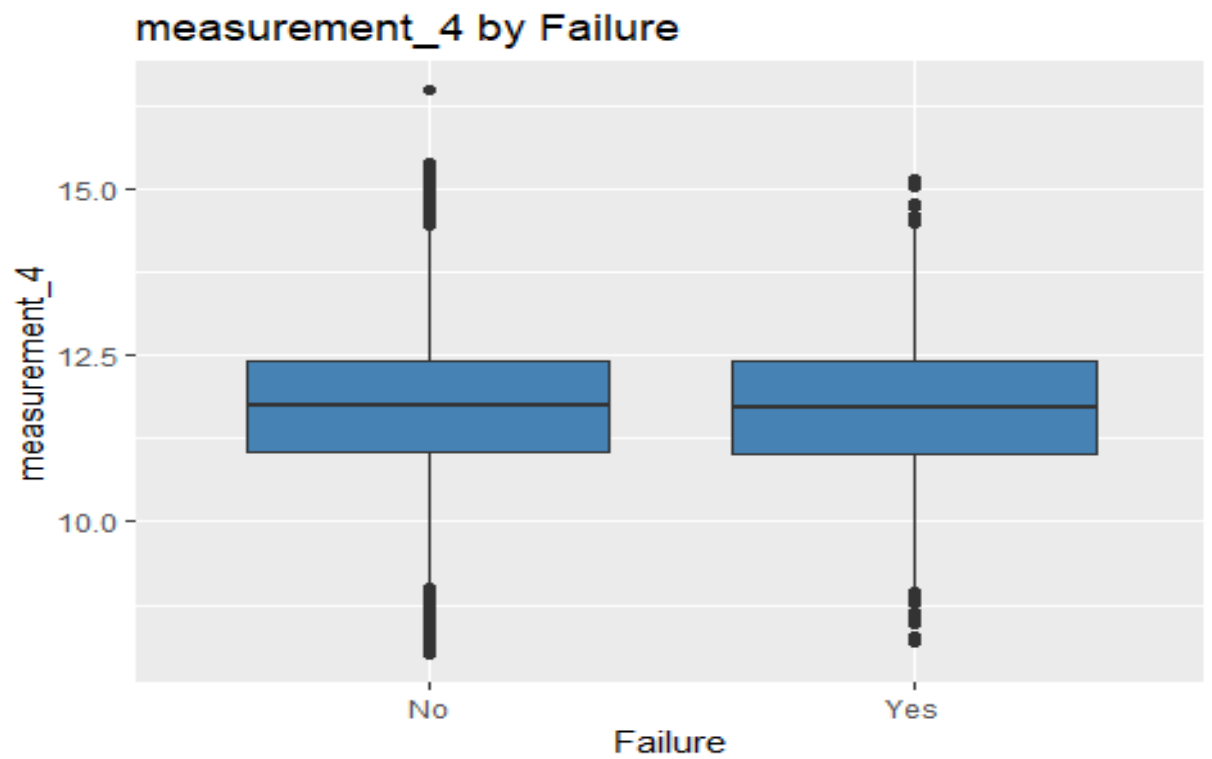




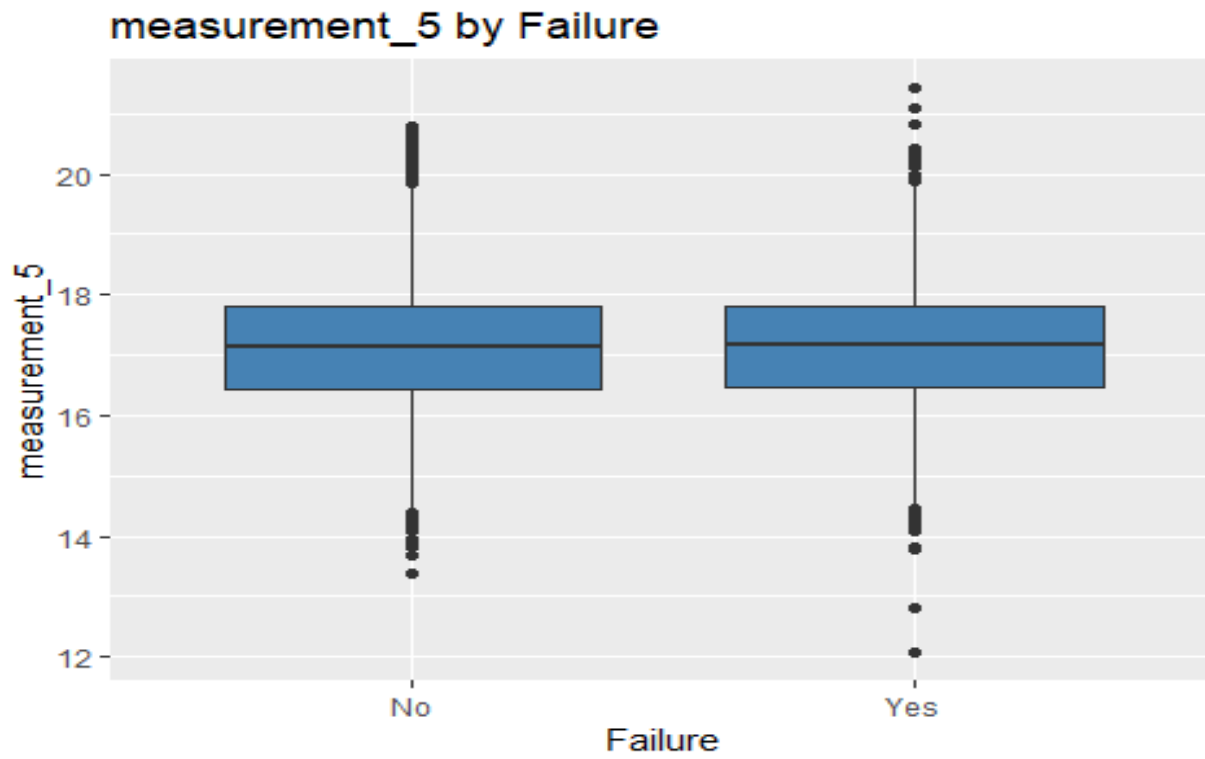
```
## Warning: Removed 381 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

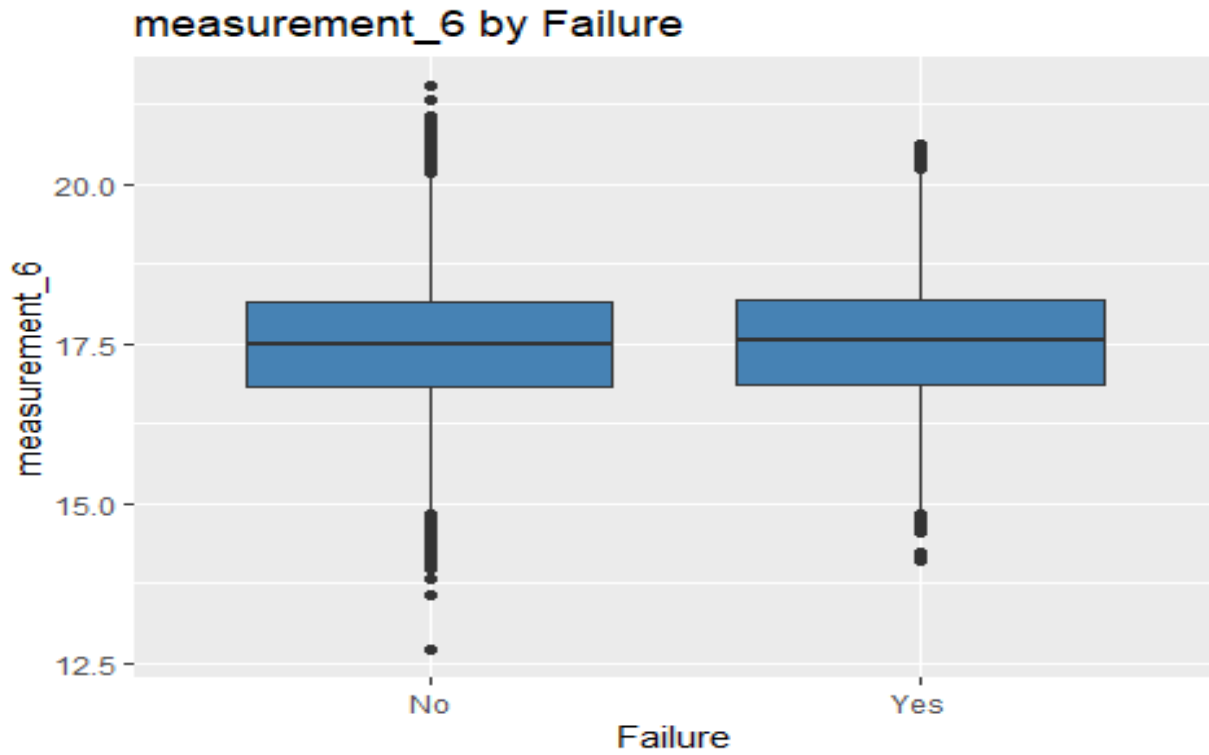
```
## Warning: Removed 538 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



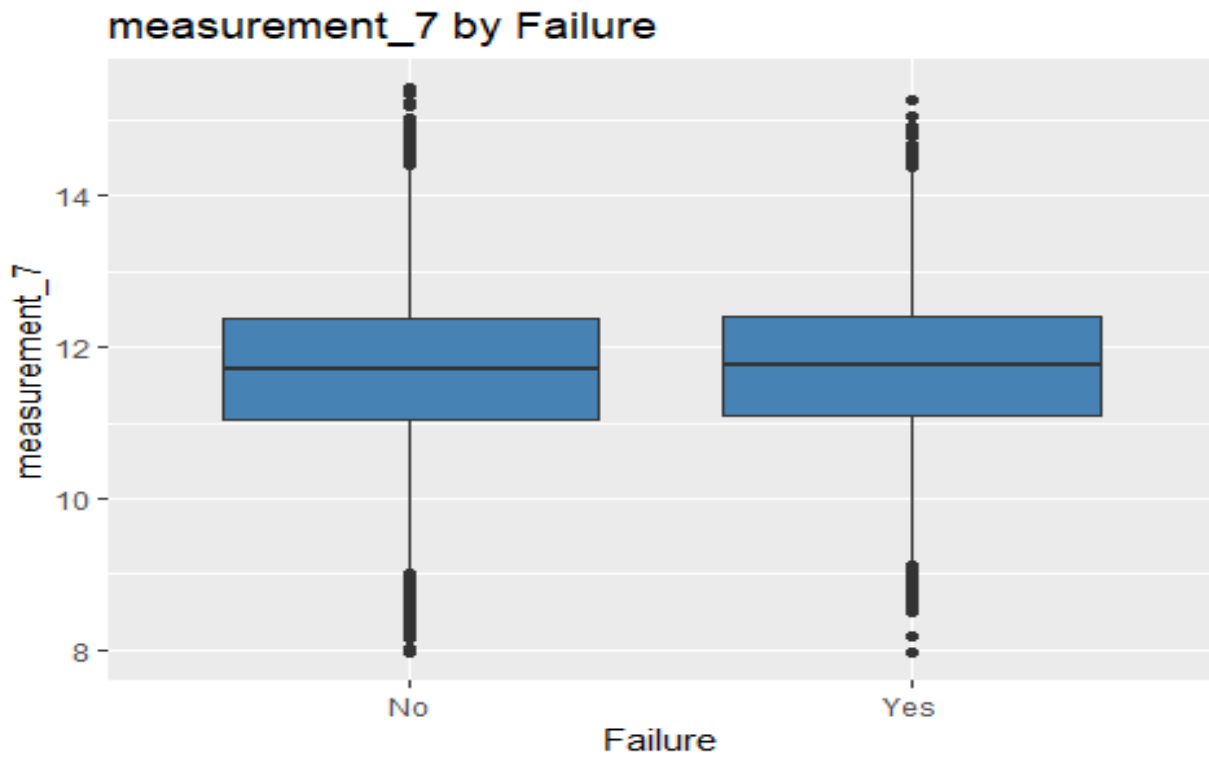
```
## Warning: Removed 676 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



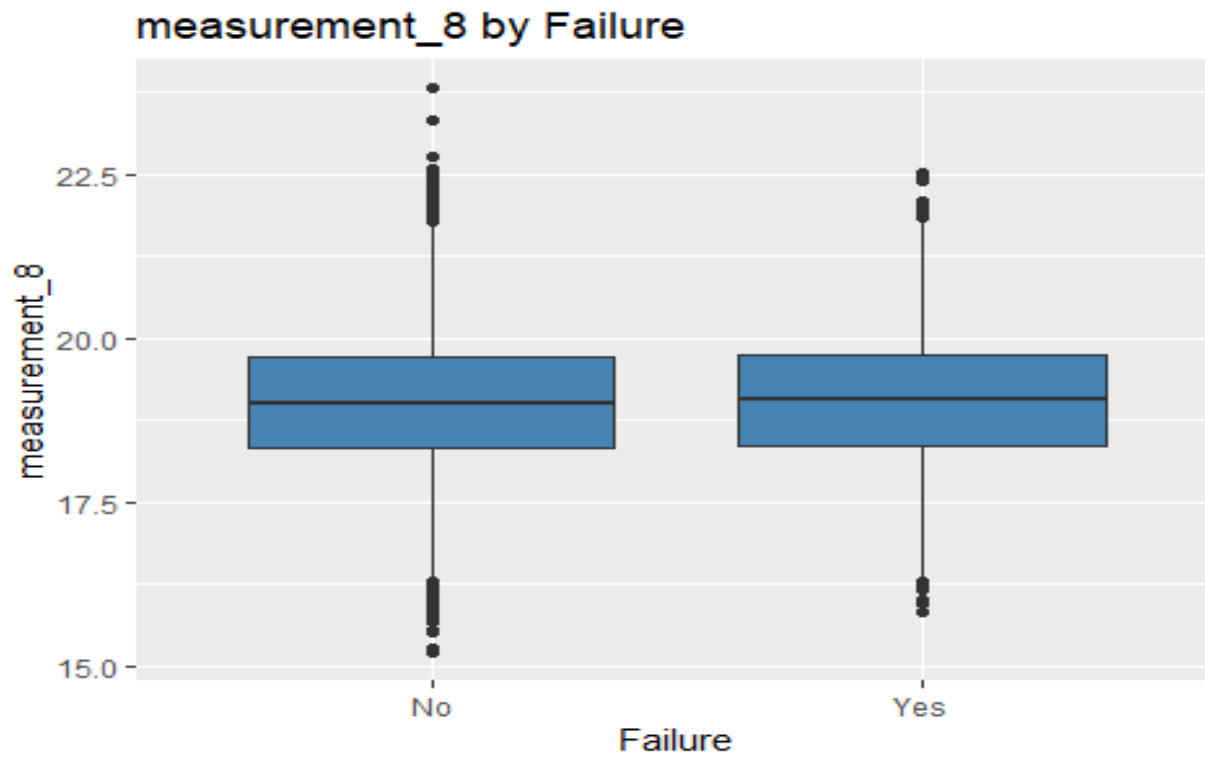
```
## Warning: Removed 796 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



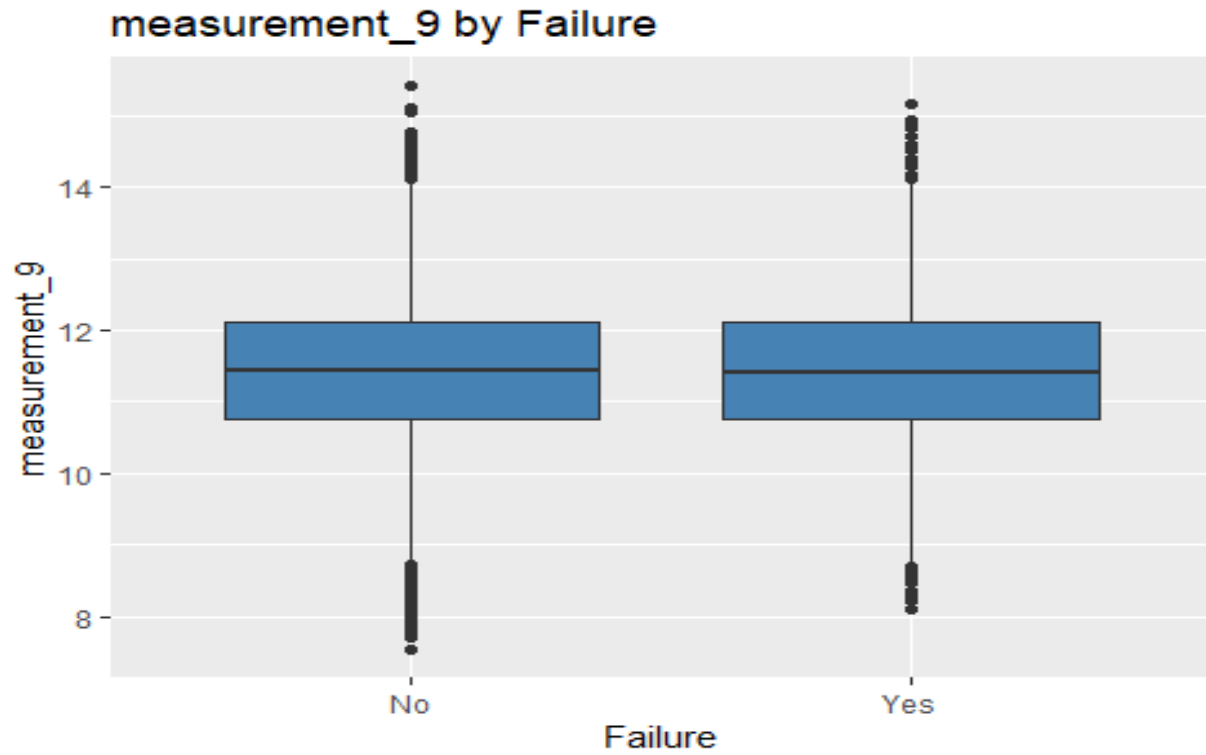
```
## Warning: Removed 937 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



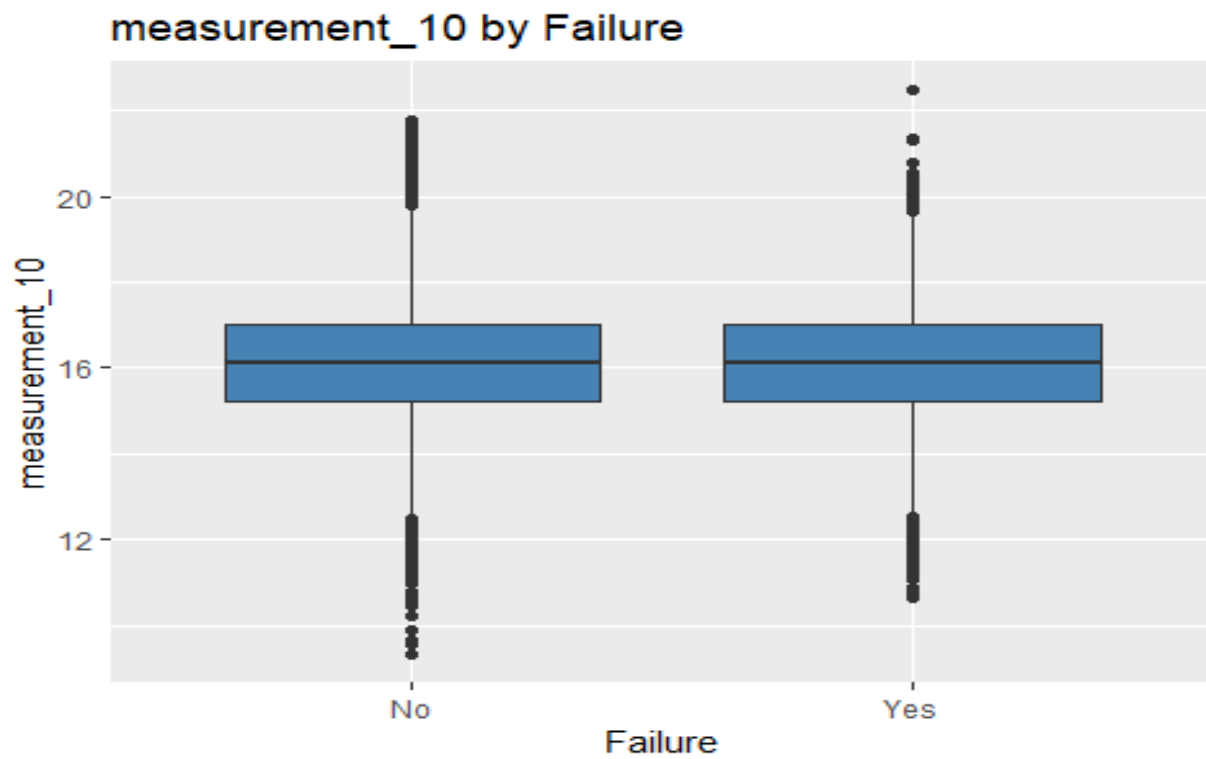
```
## Warning: Removed 1048 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



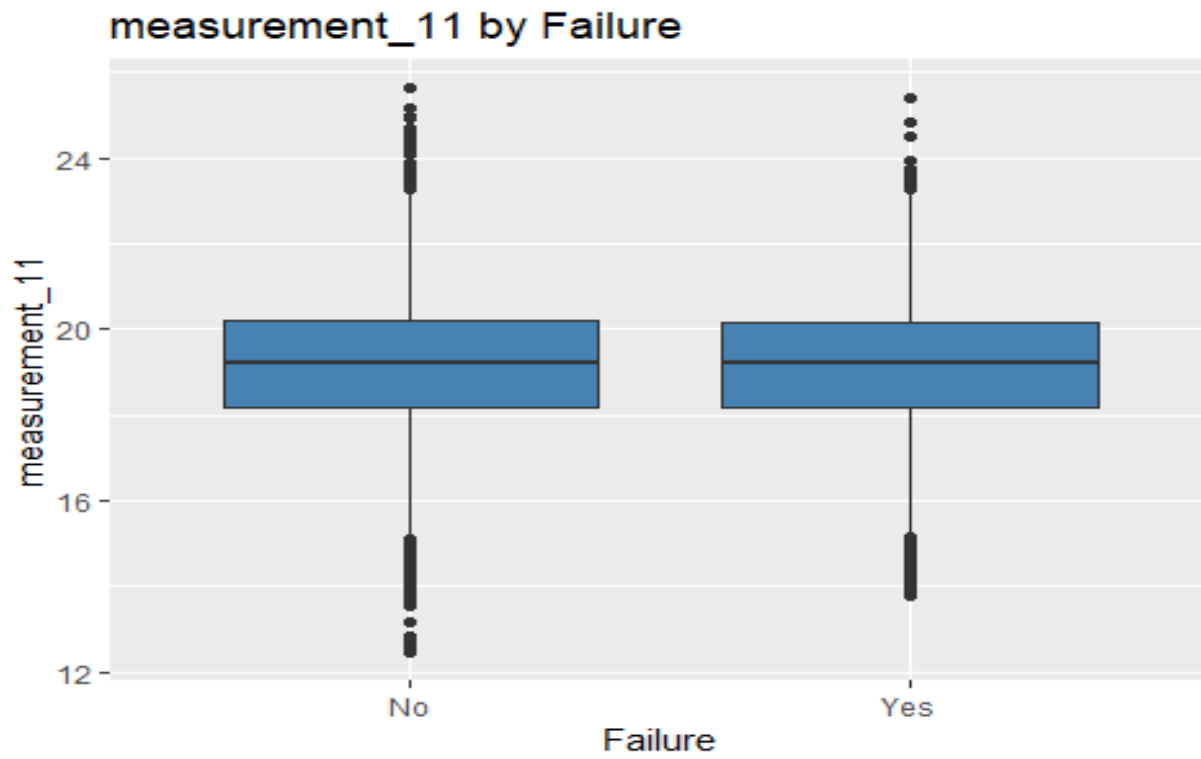
```
## Warning: Removed 1227 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



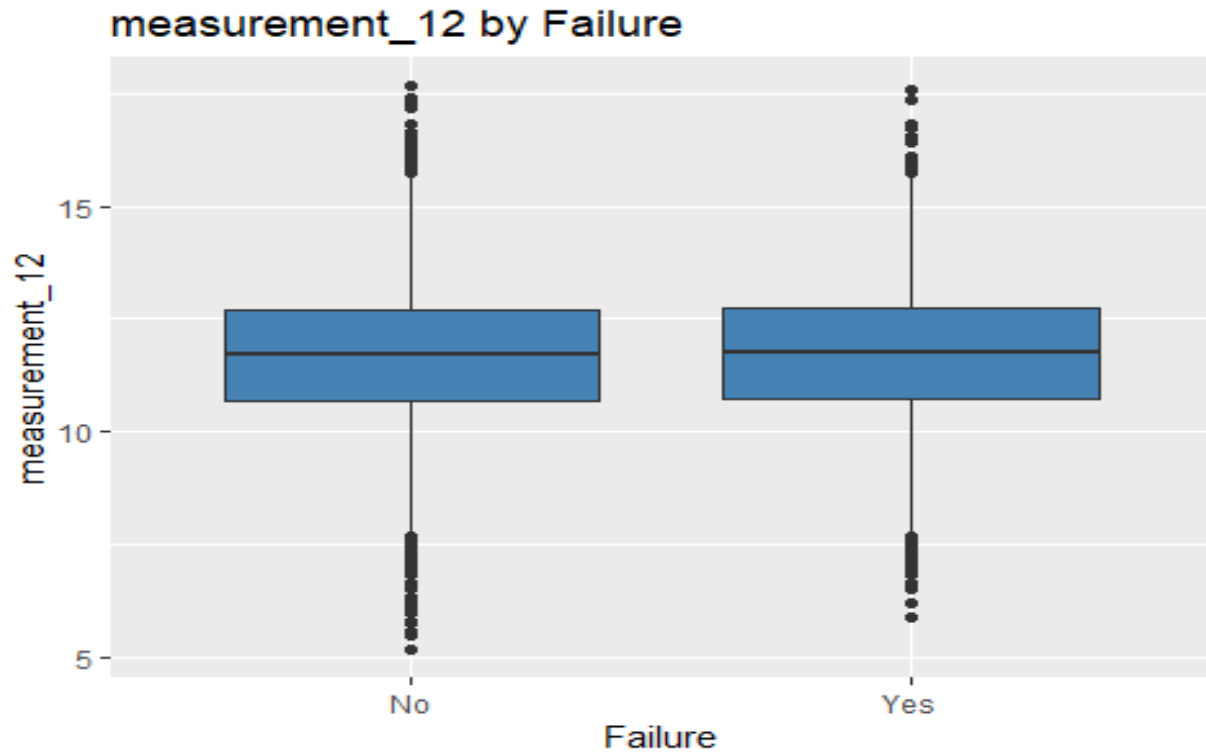
```
## Warning: Removed 1300 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



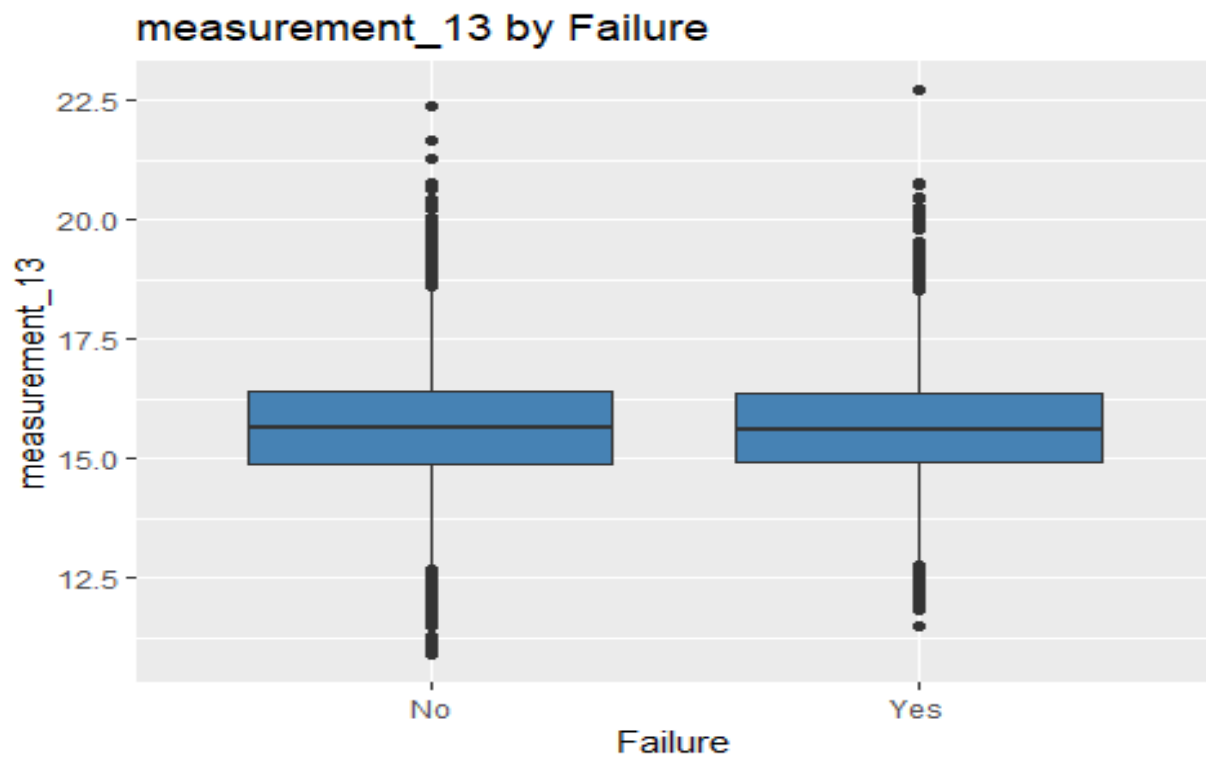
```
## Warning: Removed 1468 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



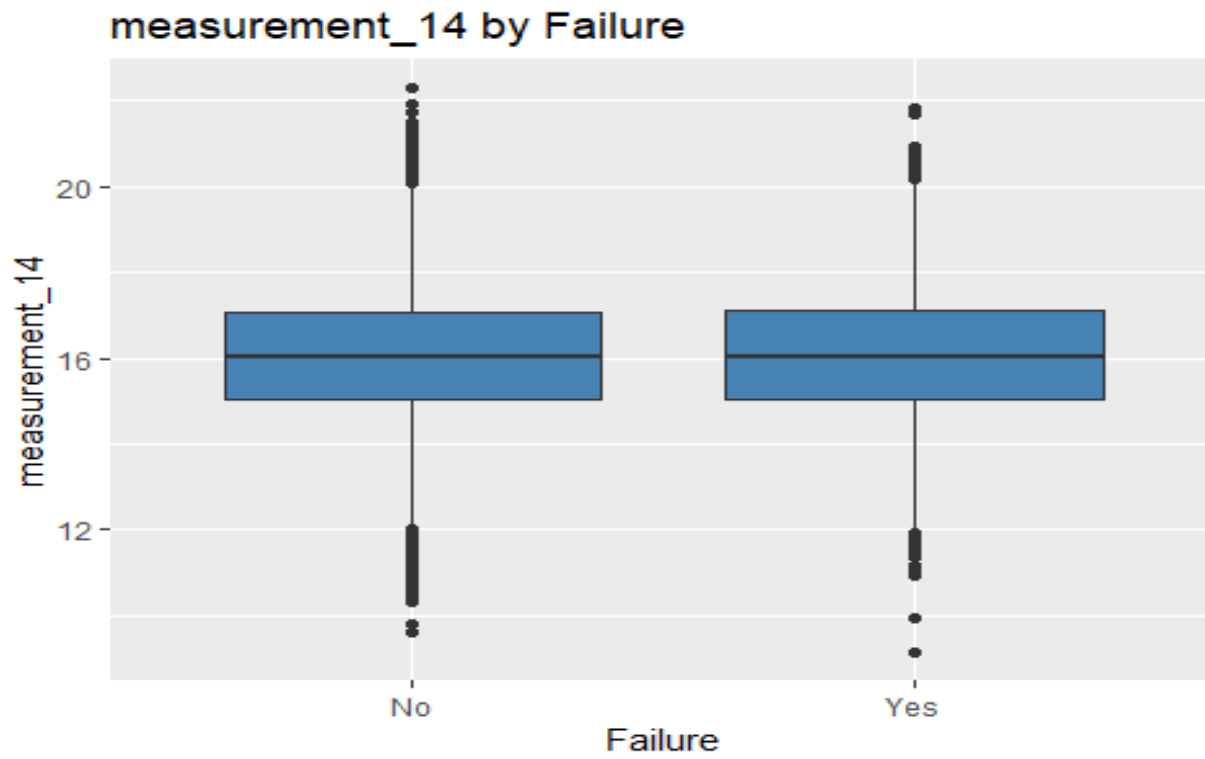
```
## Warning: Removed 1601 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



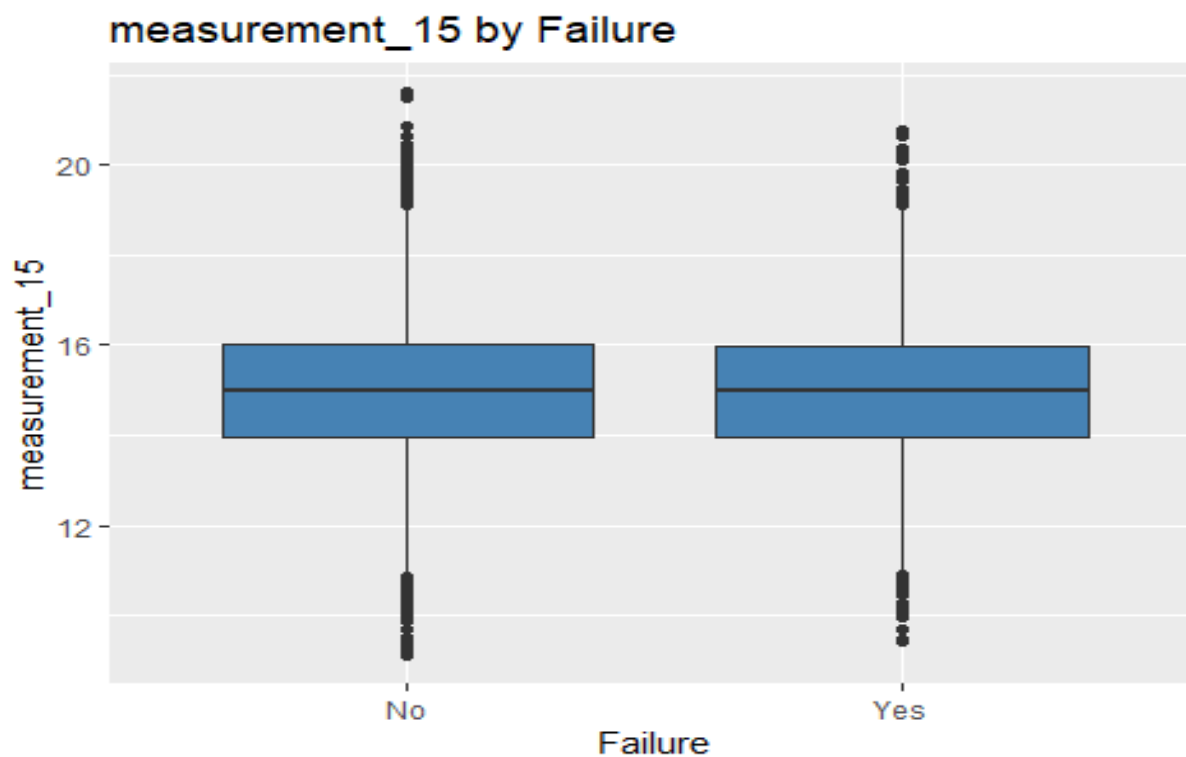
```
## Warning: Removed 1774 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



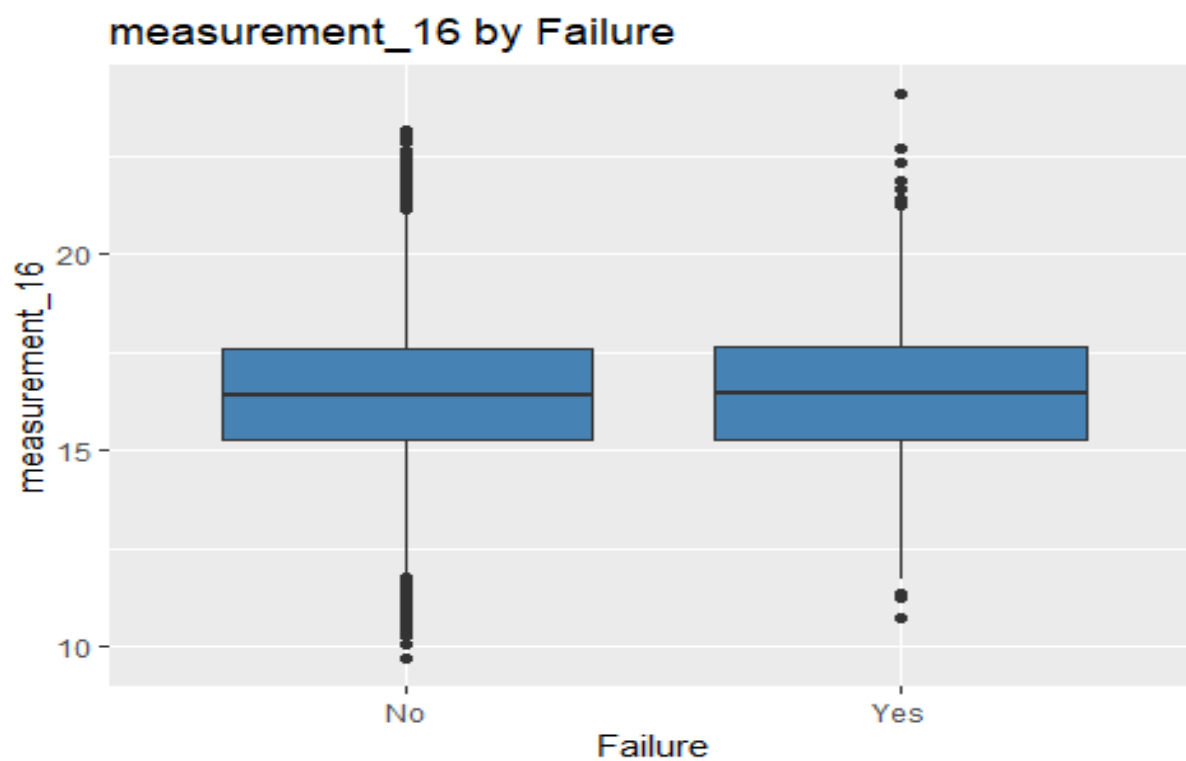
```
## Warning: Removed 1874 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



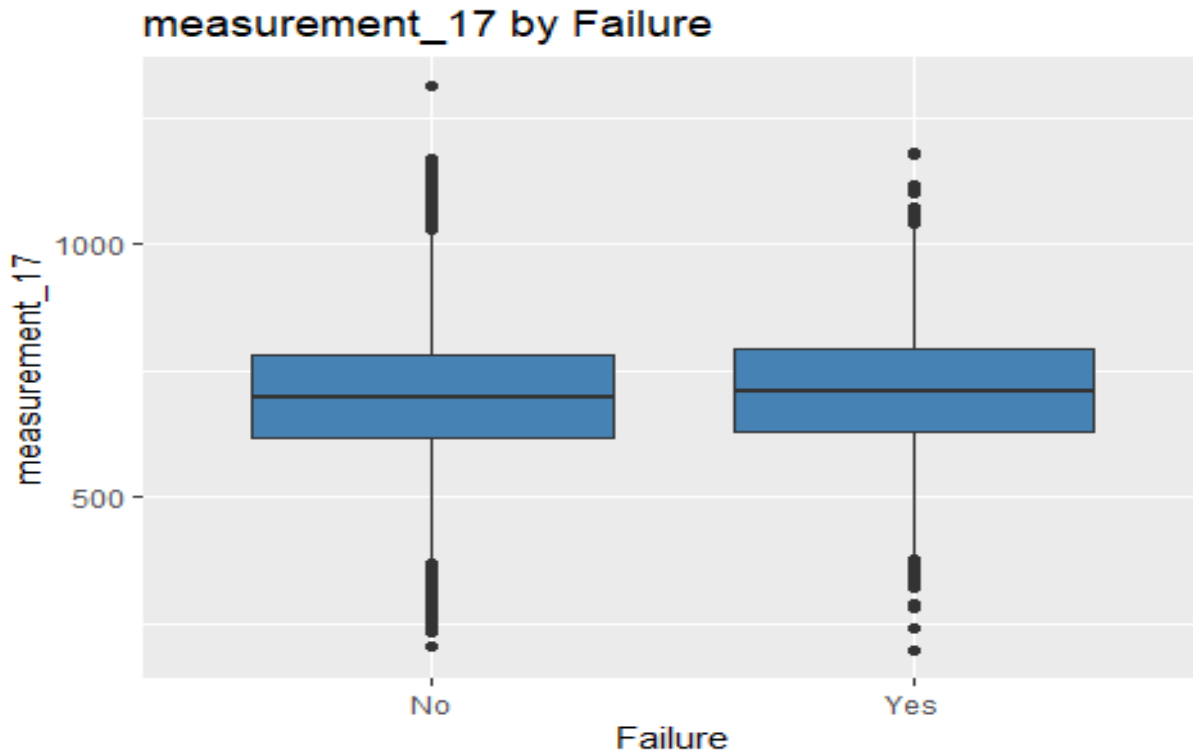
```
## Warning: Removed 2009 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 2110 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



```
## Warning: Removed 2284 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



```
# Categorical features analysis
cat_features <- df %>% select_if(is.factor)

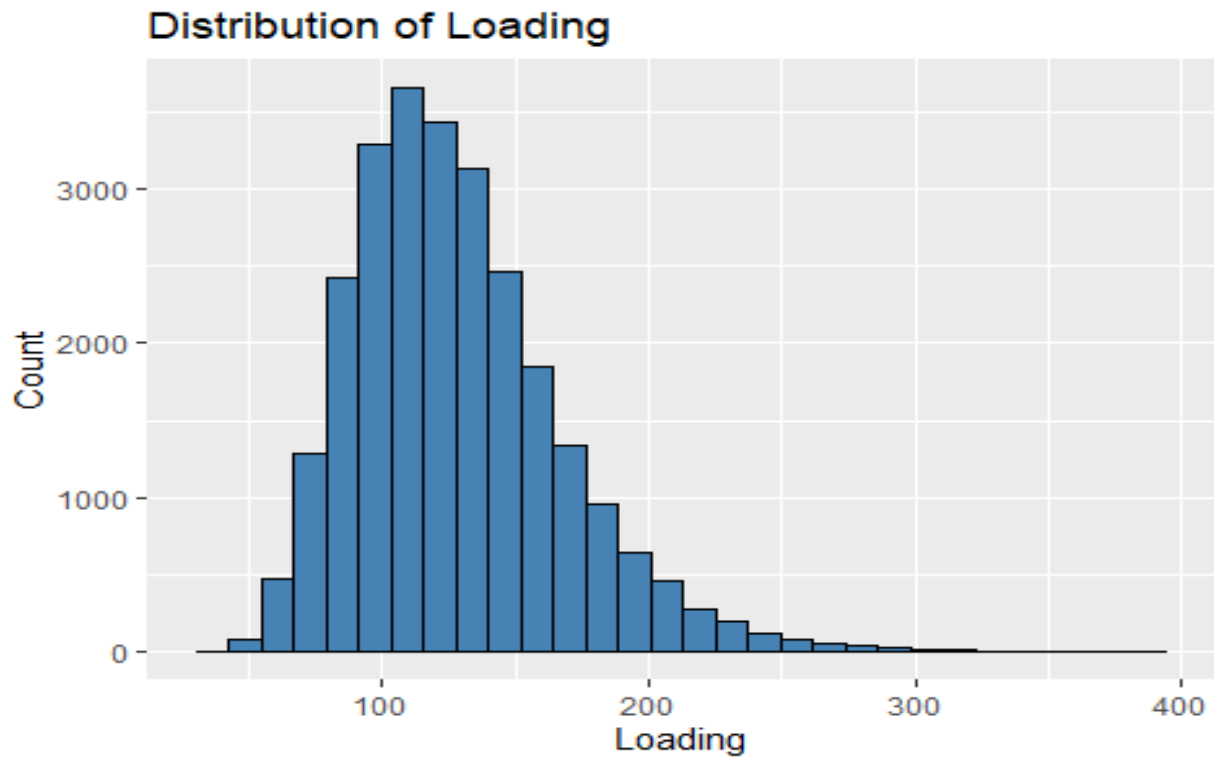
summary(cat_features)

## < table of extent 0 x 0 >

# Distribution of categorical features
for (feature in names(cat_features)) {
  p <- ggplot(df, aes_string(x = feature)) +
    geom_bar(fill = "steelblue") +
    labs(title = paste("Distribution of", feature), x = feature, y = "Count")
  +
    theme(axis.text.x = element_text(angle = 90, hjust = 1))
  print(p)
}

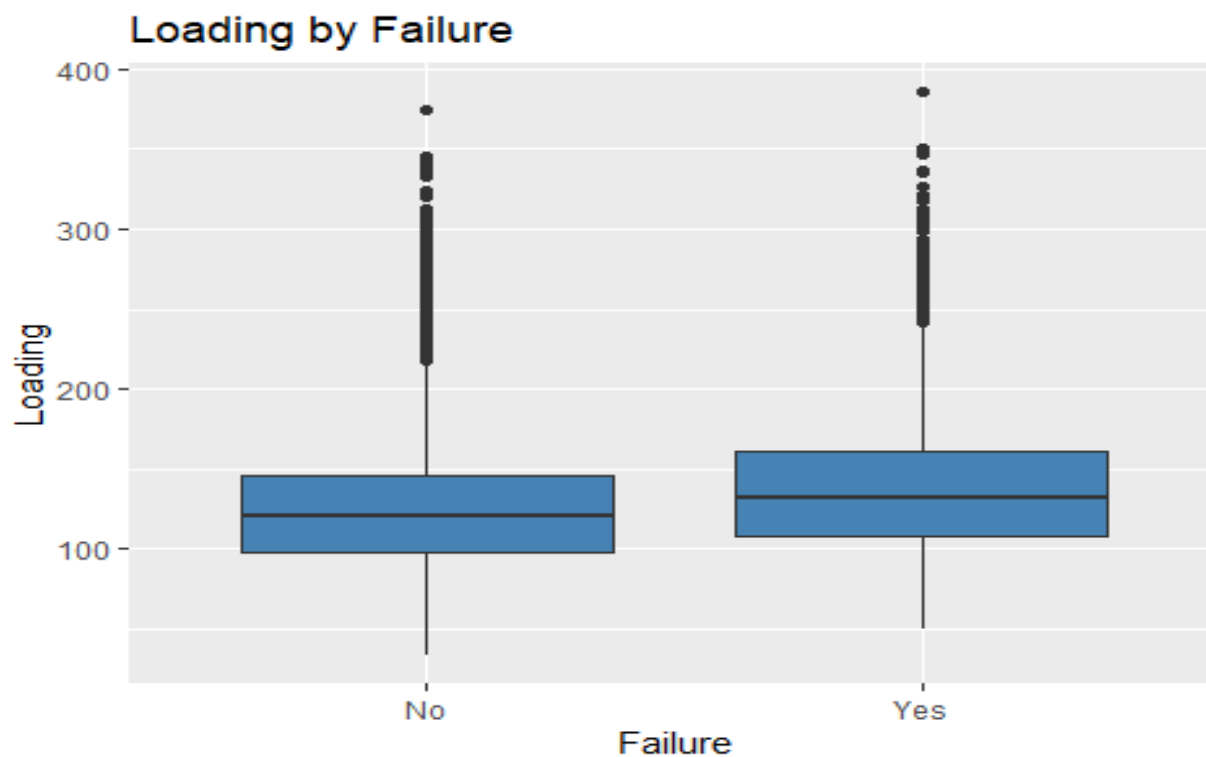
# Analysis of Loading feature
ggplot(df, aes(x = loading)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Loading", x = "Loading", y = "Count")

## Warning: Removed 250 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

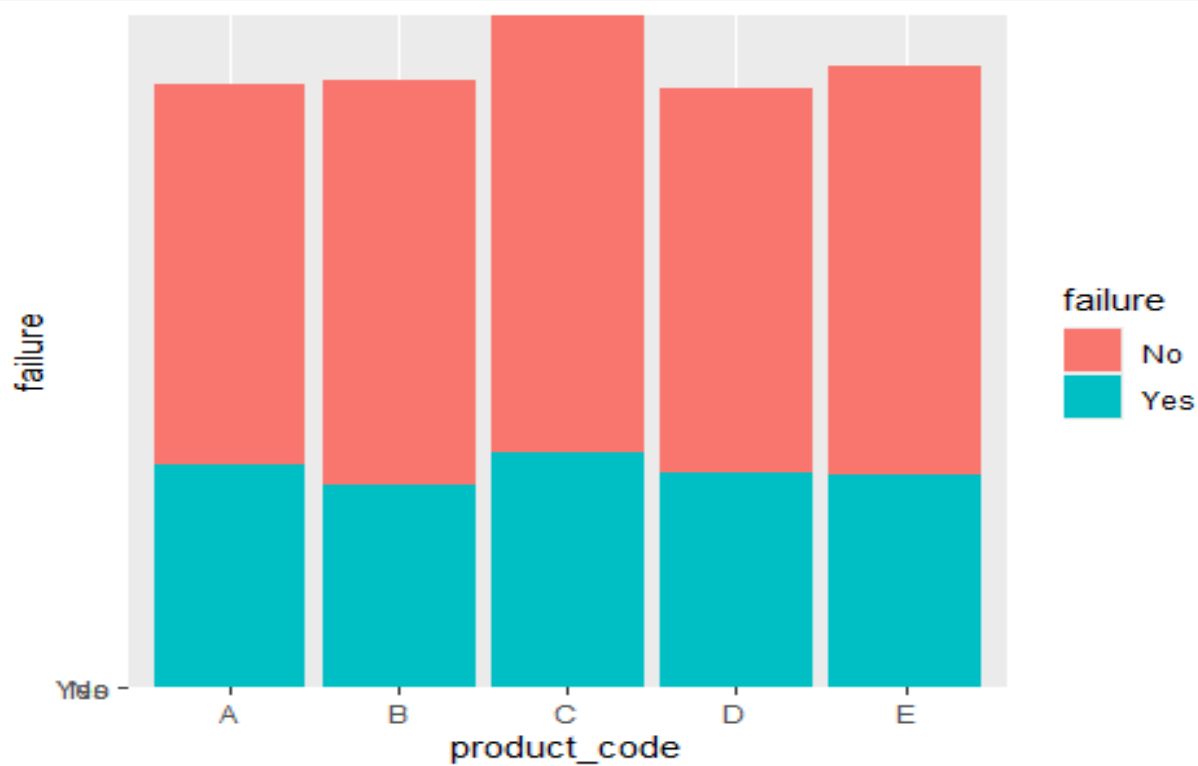


```
# Loading vs. failure
ggplot(df, aes(x = failure, y = loading)) +
  geom_boxplot(fill = "steelblue") +
  labs(title = "Loading by Failure", x = "Failure", y = "Loading")

## Warning: Removed 250 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



```
# Failure by Product Code
ggplot(df, aes(product_code, failure, fill = failure)) +
  geom_col()
```



```

# Combine Measurements
df <- df %>% pivot_longer(cols = starts_with("measurement"), names_to =
"Measurements", values_to = "Value")

# Standardize measurements
df$attribute_2 <- paste0("material_", df$attribute_2)

# Standardize measurements
df$attribute_3 <- paste0("material_", df$attribute_3)

# Combine Attributes
df <- df %>% pivot_longer(cols = starts_with("attribute"), names_to =
"Attributes", values_to = "Materials")

# ID & failure by Product code
ggplot(df, aes(id, failure, color = product_code)) +
geom_col()

```

